

Mechanisms and pathology of protein misfolding and aggregation

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Abstract

Despite advances in machine learning-based protein structure prediction, we are still far from fully understanding how proteins fold into their native conformation. The conventional notion that polypeptides fold spontaneously to their biologically active states has gradually been replaced by our understanding that cellular protein folding often requires context-dependent guidance from molecular chaperones in order to avoid misfolding. Misfolded proteins can aggregate into larger structures, such as amyloid fibrils, which perpetuate the misfolding process, creating a self-reinforcing cascade. A surge in amyloid fibril structures has deepened our comprehension of how a single polypeptide sequence can exhibit multiple amyloid conformations, known as polymorphism. The assembly of these polymorphs is not a random process but is influenced by the specific conditions and tissues in which they originate. This observation suggests that, similar to the folding of native proteins, the kinetics of pathological amyloid assembly are modulated by interactions specific to cells and tissues. Here, we review the current understanding of how intrinsic protein conformational propensities are modulated by physiological and pathological interactions in the cell to shape protein misfolding and aggregation pathology.

Sections

Introduction

Theories of protein folding and misfolding

Links between conformational stability and aggregation

Kinetic partitioning of the native state

The amyloid state and polymorphism

Kinetic principles of amyloid formation

Interactions and toxicity of amyloids

Conclusions and perspectives

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Introduction

Proteins execute a dazzling diversity of functionalities in cells, including roles in catalysis, structure, interactions and signal transduction. To achieve this array of functions, proteins fold into specific enabling structures or ensembles, for example, to bring together a stereospecific catalytic site or to achieve structural compatibility with binding partners. In the cell, protein folding fidelity is safeguarded by the protein homeostasis network, which includes molecular chaperones and degradation pathways. During ageing, the capacity to maintain protein homeostasis declines^{1,2}; a diverse group of human diseases share an underlying failure of protein folding. The consequence of misfolding can be limited to the loss of function of the misfolded protein, as is the case in cystic fibrosis^{3,4}. However, misfolded polypeptides can clump together to form larger assemblies, called aggregates. These aggregates can trigger cellular stress, leading to dysfunction and, eventually, death; that is, they mediate a toxic gain of function.

Amyloid fibrils are ordered, insoluble protein aggregates that have a direct role in many pathologies, each characterized by the proteins deposited and the tissues affected^{4,5}. They underlie many neurodegenerative disorders, such as Alzheimer disease, Parkinson disease and frontotemporal dementia^{4,5}. Amyloids can also affect other tissues, for instance in the case of type 2 diabetes mellitus, or cause systemic forms of disease, such as in the case of immunoglobulin light chain amyloidosis or transthyretin (ATTR) amyloidosis⁶ (see Supplementary Table 1). The transfer of amyloid fragments, typically referred to as 'seeds', from one organism to another is often sufficient to cause further aggregation of the normal protein in the recipient by recruitment to the growing amyloids^{7,8}. The pathological relevance of this observation remains under debate for many proteins, but has wide support in the case of the prion diseases (from proteinaceous infectious particles), which are a group of fatal, neurodegenerative diseases caused by the abnormal form of the Prion protein⁹. To add to the complexity, cells also exploit amyloid fibrils as functional entities to provide physical support, to regulate activity and as compact states in which temporary storage of protein and peptide hormones is supported prior to release or under stress conditions^{4,10,11} (Box 1).

Methodological advances have expanded our view of the biological activity of amyloids. Time-resolved and single-molecule approaches have helped in reaching a better understanding of the kinetics and thermodynamics of amyloid formation, such as the rates and pathways of aggregation and the energetics of different amyloid conformations^{12,13}. Transcriptomics and proteomics have enabled characterization of amyloid species in complex biological matrices, enabling the study of the diversity and specificity of amyloid formation^{14–16}. Information derived from high-resolution methods^{17–19} is now reaching levels that support the deduction of structure–activity relationships^{20,21}.

In this Review, we provide an update regarding the thermodynamic and kinetic principles of amyloid formation and their relation to protein folding and misfolding. We highlight key molecular drivers and discuss their relevance to end-stage amyloid structures. We discuss mechanisms of toxicity, focusing on the effect of amyloid interactions with cellular components, and the association of aggregation with healthy ageing and pathology.

Theories of protein folding and misfolding

Protein folding does not proceed by random conformational sampling²². Instead, protein sequences have been shaped by natural selection so that protein folding converges on the functional conformation²³. Protein folding reactions are often conceptually described in terms of the

so-called free-energy landscape, whereby the vertical axis corresponds to the free energy of a protein whereas the horizontal axis reflects the diversity of protein conformations, with similar conformations being situated in proximity to each other along the horizontal axis (Fig. 1a). In its simplest form, the folding landscape exhibits a smooth funnel shape. The broad upper portion of the funnel encompasses numerous diverse unfolded conformations characterized by high free energy. By contrast, the lowest point represents the distinctive native conformation, which possesses the lowest free energy. The energy landscape smoothly converges from top to bottom due to the progressive formation of native-like interactions, thereby increasingly favouring the native state of the protein. This situation corresponds to the original thermodynamic postulate²⁴ (also referred to as Anfinsen's dogma²⁵) stating that protein folding occurs spontaneously because the native conformation of a protein represents its most stable but also kinetically accessible state.

In reality, proteins often undergo folding processes involving intermediate states and encounter rate-limiting steps influenced by local structural propensities and hydrophobic collapse²⁶. Furthermore, not all folding intermediates contribute to productive protein folding, as some can misfold by forming non-native interactions^{27,28}. As a result, the energy landscapes representing protein folding exhibit a 'rugged' rather than smooth appearance (Fig. 1a). Protein misfolding also causes proteins to stray away from their original native folding trajectory. When misfolded proteins interact this eventually leads to aggregation. On protein energy landscapes this is illustrated by the presence of a 'dark side' in which proteins become trapped in energy minima-associated aggregated protein conformations. Protein folding reactions are therefore much more complex than initially anticipated by the thermodynamic postulate. Protein energy landscape theory revealed that the fundamental reason underlying the kinetic complexity of protein folding stems from structural frustration. This concept suggests that the local structural propensities of protein chains might not always align with – and sometimes even oppose – the native conformation, resulting in a rugged energy landscape with multiple local minima affecting the efficiency of protein folding. Structural frustration is probably unavoidable owing to non-aligning selective pressures for function, folding and expression levels (among others). As a result, molecular chaperones²⁹ play a vital role in guiding proteins through the folding landscape, facilitating the avoidance or resolution of unproductive conformations, thereby achieving productive folding³⁰.

An additional challenge to the original thermodynamic postulate results from protein supersaturation. Supersaturation occurs when a solute (here a protein) stays in solution for an extended period at a concentration exceeding its solubility limit at equilibrium. It has been argued that the physiological expression levels of proteins are often close to, and sometimes supersede, their solubility limit³¹. In this view, under physiological conditions the aggregated state of many proteins is thermodynamically more favourable than the native state³². Supersaturation is proposed to give rise to a metastable sub-proteome, which becomes particularly susceptible to aggregation during ageing³³ and disease^{34–36} (Box 2). Finally, proteins start to fold and assemble while still being translated on the ribosome. Co-translational folding affects the kinetics of the process, as conformational information is added as the polypeptide chain elongates, and therefore requires chaperones for proper folding^{37–39}.

Thus, although protein folding is evolutionarily shaped by thermodynamic principles, the kinetic complexities arising from structural frustration and the reliance on cellular machinery often challenge the

thermodynamic hypothesis³⁸. This point has been illustrated in detail in a study of the folding pathway of tubulin, which interacts with the chaperones prefoldin and Tric⁴⁰. In contrast to the GroEL/ES chaperone (whose inner chamber primarily provides a secluded environment for proteins to fold in^{41,42}, called an Anfinsen cage⁴³), the folding trajectory of tubulin is shaped by guiding cross-interactions with the Tric chamber, to the extent that it cannot fold independently of Tric⁴⁰.

Overall, the presence of structural frustration, supersaturation, co-translational folding and the reliance on chaperones⁴⁴ highlight the crucial role of kinetic control in cellular protein folding. This interplay of kinetic effects, pathological interactions and structural frustration in the amyloid conformation is also likely involved in amyloid polymorphism (discussed below), again resulting in a rugged energy landscape (Fig. 1a).

Links between conformational stability and aggregation

Protein architectures can be categorized in four main classes: globular soluble proteins, fibrous proteins, transmembrane proteins and intrinsically disordered proteins (IDPs). Globular proteins are stabilized by amphipathic α -helices and β -sheets that assemble to form a hydrophobic core. Often these secondary structural elements are fully buried, which is why most primary sequences harbour short hydrophobic segments (six to ten residues). These segments are essential for the stability of the hydrophobic core (Fig. 1b); however, they also constitute

aggregation-prone regions (APRs) that favour intermolecular β -sheet self-assembly into amyloid-like aggregates⁴ (Fig. 1c). APRs are found widely across proteins, regardless of their structural class^{18,45–47} (see Supplementary Table 1) and can be predicted computationally (Box 3): on average, ~20% of residues in globular domains reside within APRs.

APRs are prevalent in transmembrane proteins (Fig. 1b), due to the high hydrophobicity of their transmembrane-spanning domain. Conversely, given that the absence of cooperative folding stems from their lack of sufficient hydrophobic amino acids, the number of APRs in IDPs is substantially lower⁴⁸. However, many IDPs still require a degree of local structure propensity⁴⁹, often at interaction sites requiring some degree of hydrophobicity, and hence harbour APRs^{50–52} (Fig. 1b). Owing to their increased exposure, APRs in disordered regions of proteins are particularly susceptible to self-assembly⁵³, which probably explains why several proteins at the centre of major amyloidosis diseases – such as the amyloid- β peptide (A β) and tau for Alzheimer disease, and α -synuclein (α S) for Parkinson disease – are intrinsically disordered (see Supplementary Table 1). Molecular dynamics simulations indicate that the conformational conversion of IDPs towards an organized structure through aggregation or upon binding to functional partners follows a rugged energy landscape that is not dictated by a global minimum, such as in the case of folded proteins, and makes them prone to parallel oligomerization pathways and amyloid formation⁵⁴. Interconversion from an inert intrinsically disordered monomer to an aggregate-compatible compact monomer

Box 1

Functional amyloids

Functional amyloids have interesting properties that can be employed in biological systems¹⁰ (see Supplementary Table 1). For example, amyloid filaments are often used as a structural support or as protective layers^{333–337}. During hormone secretion, amyloid aggregation acts as a temporary storage of polypeptide material prior to secretion^{338,339}. Finally, amyloid aggregation is a modification of the functional state of a protein that can be employed to encode molecular memory through faithful templating, such as in the case of the yeast prions³⁴⁰, and also in terms of synaptic memory, as shown in the *Drosophila* brain³⁴¹.

Early structural evidence indicated that functional amyloids form continuous 'cross- β '-like structures such as β -helices or β -solenoids, thus diversifying structurally from their pathological counterparts³⁴². The repetitive nature of sequences involved in the formation of functional amyloids was proposed to support this conformation, with successive rungs corresponding to individual or sets of repeats that are capped at their ends^{342–346}. Early NMR-derived structures of the HET-s prion validated this notion³⁴¹, similar to cryo electron microscopy (cryo-EM) results and previous modelling work revealing that the major curli subunit (which is the primary scaffold for the formation of bacterial biofilms) adopts a canonical β -solenoid fold^{347,348}. However, more recent structural findings have shown that functional amyloid scaffolds are much more complex than previously considered. This complexity is particularly evident for short sequence segments. For example, the necrosome complex is composed of heterotypic amyloid fibrils that are formed by the homologous RIPK1

and RIPK3 proteins and contain an unusual ladder of alternating serines and cysteines²³⁶. Antimicrobial sequences expressed by amphibians form fibrils that can conformationally switch from a typical cross- β fold to an α -helical fibril arrangement when found in the presence of membranes³⁰⁰.

Functional amyloids are also primarily characterized by largely hydrophilic cores, in contrast to the majority of pathological amyloids³⁴⁹. The hnRNPDL ribonucleoprotein (involved in transcription and RNA-processing) forms non-toxic fibrils that bind nucleic acids³⁵⁰; their core consists of highly hydrophilic residues incorporating several water channels. The low complexity domain of hnRNPA2 also forms fibrils with a hydrophilic core but further introduces backbone kinks that facilitate the formation of reversible hydrogels containing hnRNPA2 fibrils³⁵¹. Orb2, a synaptic translation regulator involved in memory, forms amyloid folds with a core consisting of glutamines and histidines in a pH-sensitive mechanism, suggesting that these fibrils might also have a dynamic nature³⁵². This reversible property extends to β -solenoids, as in the case of β -endorphin fibrils³⁵³, which are stored in acidic secretory granules and retain a glutamate residue in a critical protonated state in their core; on exocytosis, the pH change promotes the release of the peptide hormone in the blood in a stepwise manner³⁵⁴. Together, these findings suggest that functional amyloids utilize atypical amyloid folds that promote the stacking of polar and charged side chains to promote their functionality and reversibility, in contrast to the rigid hydrophobic packing of their pathological counterparts³⁴⁹.

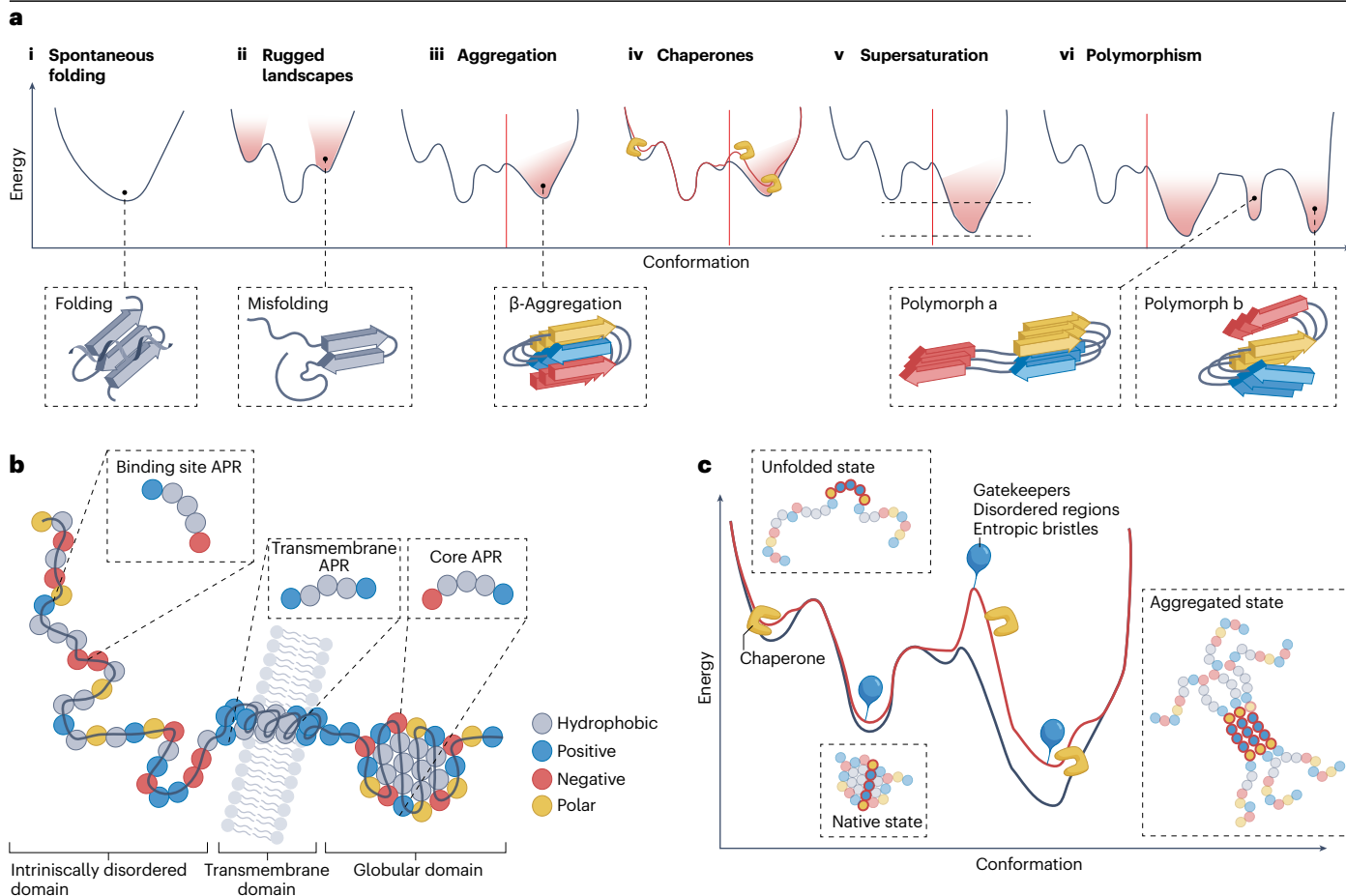


Fig. 1 | The role of frustration and aggregation-prone regions in protein folding and aggregation. **a**, Protein folding reactions are often conceptually described in terms of the so-called free-energy landscape, whereby the vertical axis corresponds to the free energy of a protein whereas the horizontal axis reflects the diversity of protein conformations. In its simplest form, the folding landscape exhibits a smooth funnel shape (panel i). The energy landscape smoothly converges from top to bottom due to the progressive formation of native-like interactions, thereby increasingly favouring the native state of the protein. In reality, proteins often undergo folding processes involving intermediate states and encounter rate-limiting steps. As a result, the energy landscapes representing protein folding exhibit a 'rugged' rather than smooth appearance (panel ii). Protein misfolding also causes proteins to stray away from their original native folding trajectory, illustrated by the presence of a side of the energy funnel where proteins become trapped in energy minima associated with aggregated protein conformations (panel iii). Molecular chaperones play a vital role in guiding proteins through the folding landscape, facilitating the avoidance or resolution of unproductive conformations (panel iv). An additional challenge

is protein supersaturation. Under physiological conditions, the aggregated state of many proteins is thermodynamically more favourable than the native state³² (panel v). The interplay of kinetic effects, pathological interactions and structural frustration is also likely involved in amyloid polymorphism, again resulting in a rugged energy landscape (panel vi). **b**, Essential states of functional proteins in cells. Owing to structural and functional requirements, ~20% of natural protein sequences comprise hydrophobic segments (fewer in intrinsically disordered proteins (IDPs)). Following this principle, aggregation-prone regions (APRs) are integral factors that facilitate the folding of soluble protein domains and frequently occur in transmembrane domains or as binding surfaces in IDPs. These segments generally adopt a regular secondary structure in native structural ensembles. **c**, Negative selection and chaperones provide kinetic control over APRs. Natural selection favours native-like folding by minimizing the effect of APRs. Elements of negative design opposing aggregation include gatekeeper residues or entropic bristles. Structural frustration due to APRs is controlled by both negative selection and chaperones.

state has been proposed to occur for multiple IDPs^{55–57}, and lipid interactions with disordered monomers might also progress to pathology⁵⁸. Similar interactions have been suggested to lead to membrane disruption induced by amyloid-like aggregates formed by IDPs⁵⁹. This rapid interconversion and compaction of IDPs might be countered by interactions with solvent molecules that counter-intuitively retain IDPs in an expanded conformation in aqueous solutions⁶⁰. Finally, transient misfolding events occur frequently between tandem repeats

of neighbouring domain copies incorporated in fibrous proteins⁶¹. Neighbouring domains sharing high sequence similarity are more prone to amorphous aggregation^{61,62} because they can expose identical APRs in proximity, thus increasing their susceptibility to aggregate, exchange identical segments between domains or form self-recognition contacts that introduce frustration in their energy landscape and entropically favour misfolding compared with native folding⁶³.

The determining role of APRs for protein aggregation has been recapitulated by many studies, starting with mutational analysis^{46,64,65} and grafting experiments, in which cloning of an APR derived from an aggregation-prone protein into another, typically non-amyloidogenic sequence is enough to induce amyloid formation^{66,67}. Several studies have demonstrated their ability to assemble in isolation⁴⁵, as well as their ability to induce aggregation of the full-length protein when added as peptides in *cis* formation^{68–71}. As such, APRs probably constitute the most common occurrence of structural frustration in proteins whereby local propensity for amyloid-like aggregation competes with and opposes global native structure.

Despite being a result of protein architecture, the intrinsic tendency of proteins to aggregate, particularly through APRs, poses challenges to cellular proteostasis and contributes to disease. So why, then, has aggregation not been eliminated by natural selection? One reason is that mutations that destabilize the native state of proteins often lead to increased aggregation propensity⁷². Proteins from extremophiles provide an illustrative example of the seemingly paradoxical nature of APRs: although these proteins exhibit increased resistance to thermal denaturation, they harbour more (rather than fewer) APRs owing to their more hydrophobic cores^{73,74}. Conversely, the selective pressure of Hsp90 deletion on proteins of Hsp90-dependent viruses results in less aggregation-prone, but also less stable, proteins⁷⁵. These findings demonstrate the intricate relationship between protein aggregation and stability. Moreover, native and amyloid structures

are thermodynamically correlated, as mutations that destabilize the amyloid state often also destabilize the native structure^{74,76}. Secondly, the universal genetic code is conservative with regard to both protein structure and aggregation propensity, as many single base pair substitutions result in conservative substitutions that preserve protein stability⁷⁷, but also tend to conserve protein aggregation^{74,78}.

An alternative thought-provoking perspective suggests that amyloid propensity in present-day proteins may have its roots in ancient protein structures. Indeed, the amyloid-first hypothesis proposes that prebiotic amyloid peptides gave rise to the universe of current protein topologies^{79,80}. In this view, regions with amyloid propensity persist in contemporary proteins as remnants of this primordial influence. Interestingly, amyloid peptides have demonstrated the ability to spontaneously emerge⁸¹ and propagate⁸² in prebiotic conditions, bind RNA⁸³, interact with lipid membranes⁸⁴ and harbour catalytic function⁸⁵.

Kinetic partitioning of the native state

The native state of most proteins is a metastable state that is protected from misfolded and aggregated states through kinetic partitioning (Fig. 1a,c). For proper function, the rate of folding to the native state must be higher than the rate of aggregation at physiological concentrations, and the rate of unfolding should be sufficiently slow so that the native state remains stable throughout the biological lifetime of the protein. From an energy landscape perspective, the barriers between the native state and aggregated states need to be sufficiently high

Box 2

Supersaturation

Supersaturation refers to a concentration of a solute that exceeds its solubility limit, which is defined as its concentration at equilibrium in the same given conditions. This idea extends to parts of the proteome expressed at levels beyond their solubility, a condition that sets these proteins at high risk of aggregation and requires strict proteostatic control³⁵. High local concentrations have been suggested to shift Anfinsen's postulate from favouring intramolecular interactions and natural folding towards conditions promoting intermolecular contacts and assembly³⁵⁵. Aggregation-prone proteins, for which the solubility barrier is lower, are particularly susceptible to this shift^{355,356}. The environmental context also influences the levels of protein susceptibility, as conditions reducing the solubility barrier of proteins — such as membrane interactions or charge neutralization — can promote their assembly in a salting out-like process⁵⁶. Regardless, supersaturation as a general mechanism has been suggested to drive massive aggregation and increased cellular vulnerability in major neurodegenerative disorders^{35,357}. In fact, specific pathways enriched in supersaturated proteins are downregulated in Alzheimer disease brains to mitigate massive subproteomic collapse in conditions of compromised proteostasis³⁵⁸. Supporting this finding, single-cell transcriptomics and subcellular proteomics uncovered an enrichment of metastable proteins related to synaptic function and mitochondrial energy metabolism in protein inclusions related to neurodegeneration, thus providing a traceable pattern for selective vulnerability of neurons to aggregates³⁵⁹.

Proteome analyses of the composition of inclusions derived from patients with myopathy revealed a similar metastable proteome that might be linked to muscle motor dysfunction³⁶⁰. The importance of the protein quality control machinery in keeping such metastable subpopulations in line is undeniable. This premise questions, however, whether supersaturation is a driving force of ageing or vice versa. Are proteins that are expressed beyond their solubility a threat during early life when protein quality control is not yet compromised? There are arguments to be made to counter this notion. For instance, mass spectrometry data in *Caenorhabditis elegans* revealed that most proteins are expressed above their limits of solubility and that their monitoring is lost as a result of the functional decline of the machinery due to ageing³⁶¹. Gene co-expression analysis identified the endosomal-lysosomal and ubiquitin-proteasome systems as key pathways promoting clearance and preventing massive proteostatic collapse in Alzheimer disease³⁶². On the other hand, it is also possible that supersaturation drives cell ageing by compromising proteostasis. Impairment of certain molecular chaperones exacerbated the widespread collapse of a metastable proteome induced when targeting the activity of ATM and DNA topoisomerases³⁶³. In line with this, loss of chaperone-mediated autophagy in mice was shown to alter neuronal function by synergistically promoting collapse of a metastable proteome and increasing cellular vulnerability to aggregation toxicity⁹⁴.

Box 3

The impact of artificial intelligence

Significant advances have been made in protein structure prediction with the introduction of state-of-the-art artificial intelligence-based methods. One notable milestone was the arrival of AlphaFold, which employs established structural knowledge from the Protein Data Bank (PDB) and co-evolutionary sequence analysis derived from multiple-sequence alignments³⁶⁴. Through training a deep neural network to predict pairwise distances between residues in random sequences, AlphaFold successfully unveiled the structural folds of 98.5% of human proteins, including novel folds not previously observed³⁶⁴. The efficacy of AlphaFold was independently validated by a following collaborative study³⁶⁵ and although its performance is less successful for dynamic regions, other studies demonstrated that when combined with NMR or molecular dynamics-derived parameters, it can be utilized to predict protein dynamics and flexibility^{366,367}. Moreover, when integrated with experimental data from crystallography or cryo electron microscopy (cryo-EM), AlphaFold substantially improved protein modelling quality and coverage³⁶⁸. Additional methods, such as RoseTTAFold³⁶⁹, ColabFold³⁷⁰ and trRosettaX-Single³⁷¹, have emerged to enhance user accessibility, speed and accuracy of predictions for protein folding and complex formation. There has also been a focus on expanding structure predictions to macromolecular protein complexes^{372,373}, although accurate predictions for asymmetric complexes remain challenging³⁷⁴. Furthermore, ongoing efforts aim to predict protein-cofactor interactions³⁷⁵ and nucleic acid conformations³⁷⁶. Lastly, large language models, such as ProGen³⁷⁷, offer substantial benefits

to the field of protein design by generating new sequences with predictable functionality, thereby showcasing the potential for further advancements.

Despite the advances made in predicting protein folding and function, the daunting task of predicting protein aggregation remains exceptionally challenging, even for artificial intelligence-based methods³⁷⁸. Protein aggregation defies the influence of evolutionary pressure that shapes sequences, which is the foundation of such sequence-based methods. Additionally, the morphological adaptability of amyloids surpasses Anfinsen's dogma, which states that a particular structure is encoded by a single sequence.

The example of AlphaFold clearly demonstrates that achieving reliable prediction capabilities necessitates a much larger ensemble of structural information. Despite the remarkable progress made by cryo-EM and solid-state NMR in determining the structure of fibril aggregates, we are still some distance away from attaining such levels of prediction accuracy. However, the increased availability of amyloid structures has spurred the development of computational approaches that focus on analysing the stability and factors that govern protein aggregation^{18,20,21}. In this light, machine learning has made considerable advances in predicting the propensity for aggregation based on the protein sequence^{52,379,380}, as well as for the breakdown of aggregation kinetics³⁸¹. Additionally, machine learning has shown promise in the field of medical diagnosis and amyloid disease classification^{382–384}.

(Fig. 1c). In fact, proteins with a short life span are enriched in protein deposition diseases⁷² as their turnover becomes dysregulated with age due to a decline in protein degradation⁸⁶. Kinetic partitioning is achieved by both protein intrinsic and extrinsic factors.

Extrinsic factors

The most well-known extrinsic factors are molecular chaperones⁸⁷, which favour native protein folding, prevent misfolding, mediate degradation and reverse aggregation. A full overview is beyond the scope of this Review, but we mention a few highlights. For example, the different client-binding mechanisms of the Hsp70 family allow for the adaptable functionality of this hub chaperone, which can ATP-dependently stabilize or unfold protein structures and prevent aggregation by safeguarding both partially folded and unfolded protein chains⁸⁸. This plasticity is further regulated by co-chaperone activity, such as with Hsp40 (ref. 89). Another example of this cooperativity is in α S, for which six divergent chaperones recognize a motif around Tyr39; notably, phosphorylation abrogates their activity, providing a potential link for this post-translational modification in Parkinson disease⁹⁰. Chaperone-mediated native folding can be catalysed even under non-equilibrium conditions through ATP consumption⁹¹.

Chaperone-mediated autophagy is another important pathway contributing to degradation of misfolded and aggregated species⁹². For example, DJ-1 and Hsc70 mediate autophagy of α S⁹³ and other

metastable neuronal proteins that are at risk of aggregation⁹⁴. Tau acetylation reroutes degradation towards macroautophagy and endosomal microautophagy but also enhances tau prion-like transmission, which is considered important for the spreading of the pathology through the brain⁹⁵. Chaperones also differentially interact with protein aggregate species formed during the aggregation process⁹⁶ to block or disassemble protein aggregates⁹⁷. Single-particle analysis of ClpB-mediated disaggregation revealed that it is a dynamic and fast process, with substrates refolding after exiting the pore at rates of 500 residues per second⁹⁸. However, chaperone-mediated amyloid disassembly can also backfire if amyloid fragments (seeds) rather than monomers are produced; each fragment can start a new aggregation reaction (in a process called seeding, see below), as observed in the case of Hsp70 and tau amyloid fibrils⁹⁹ or during the cooperative action of DNAJB1 and Hsp70 in disassembling α S fibrils¹⁰⁰.

The other external factor affecting kinetic partitioning of the native state is translation dynamics¹⁰¹, in which variations in tRNA abundance and compatibility with the ribosome binding site lead to differential translation speeds for different codons^{102,103}. Because the genetic code is redundant (that is, multiple codons code for the same amino acid), the same sequence can be translated at different rates depending on the codons¹⁰³. Because translation is slower than protein folding by several orders of magnitude, an idea gaining traction is that the non-uniform usage of codons in organisms (bias) is an adaptation

to optimize the translation rate in favour of native folding^{104,105}. Evidence for this idea includes the following: hydrophobic cores (thus also APRs) tend to be enriched in optimal codons¹⁰⁶; there seems to be a bias in secondary structural preference of codon usage¹⁰⁷; ribosomal pausing sites in interdomain regions are proposed to enable folding between domains¹⁰⁴; and ribosomal pausing allows time for chaperone binding¹⁰⁸. Codon bias is thought to be one of the reasons why non-synonymous mutations can affect protein structure and function^{107,109}. Moreover, at least in yeast, ribosomal pausing becomes dysregulated with age¹¹⁰. Notably, translation dynamics probably synergizes with interactions between the nascent chain and the ribosome¹¹¹, as well as ribosome-associated chaperones, such as NAC¹¹².

Intrinsic factors

To promote kinetic partitioning, evolution has selectively enhanced the presence of aggregation gatekeepers^{113,114}, which are a distinct group of residues that counteract aggregation. Gatekeepers achieve this by employing charge repulsion (for example, aspartate, glutamic acid, arginine and lysine) or discouraging aggregate structure (proline). They are typically found at the amino and carboxy-terminal flanks of APRs at the first position at which the polypeptide chain emerges sufficiently from the hydrophobic core to accommodate placement of a charged residue¹¹⁵. Aggregation gatekeepers also occur in the middle of APRs in polypeptide sequences, effectively suppressing the local aggregation propensity and leaving only cryptic APRs¹¹⁶ (that is, weakly aggregating APRs that have a higher content of polar and charged residues^{48,52}). Cryptic APRs are more prevalent in IDPs than in globular proteins because of fewer structural constraints⁴⁸. Gatekeeper conservation correlates with the aggregation propensity of the APR, and this conservation comes at a cost to protein stability, suggesting that aggregation gatekeepers form an additional class of functional residues^{117,118}. The codon bias of aggregation gatekeepers differs from the average for the proteome; in addition, conservation of aggregation gatekeepers occurs at the codon level, not the amino acid level, suggesting a role in co-translational protein folding¹¹⁷. Negatively charged aggregation gatekeepers are the more potent suppressors of aggregation; however, their short side chains restrict their usage to the surface of globular proteins¹¹⁵. The longer, positively charged gatekeepers are more easily accommodated in the hydrophobic core but are less efficient aggregation inhibitors¹¹⁵. As a result, charged aggregation gatekeepers seem to constitute a code for chaperone interaction: several major chaperones, such as Hsp70, have a strong affinity for APRs flanked by positively charged aggregation gatekeepers¹¹⁵. Notably, DAXX, a polyD/E protein, has been identified as a chaperone with a high affinity for positive charges¹¹⁹.

The wider flanking regions surrounding APRs have additional modulatory effects as well¹²⁰. The mechanism of action of MOAG4/SERF, a known modifier of proteotoxicity^{121,122}, falls into this category. MOAG binds to negatively charged residue clusters distal from APRs on aggregation-prone proteins and expedites aggregation into inclusions, thereby limiting toxicity of the aggregation process to the cell¹²³. An α -helical region preceding the aggregation-prone polyQ track of Ataxin 7 stabilizes the latter in a helical conformation and suppresses aggregation^{64,124} (see Supplementary Table 1). The central tau³⁰⁶VQIVYK³¹¹ aggregation motif (see Supplementary Table 1) is locked in a β -hairpin-like compact structure by its N-terminal flanking region that prevents aggregation; alternative splicing or disease-related mutations alter the modulating capacity of this upstream element¹²⁵. Disordered flanking regions can have an inhibitory effect on APRs, as shown in the

case of MPS2, an amyloidogenic protein expressed by *Plasmodium falciparum*¹²⁶. Apart from inhibiting aggregation, such regions can also act as 'entropic bristles', which are intrinsically disordered or highly charged short sequences that increase sequence entropy and sweep out a large area through water interactions or random movements^{127,128}. As such, entropic bristles increase the fragility of fibrils and tune the seeding efficiency of aggregates by generating a larger number of short prion-competent fragments¹²⁹; they can also simply modify the resulting morphology of aggregates formed by APRs¹³⁰.

The amyloid state and polymorphism

With the establishment of structure determination methods for proteins in the amyloid state and the increasing number of available structures, we have gained insight into the remarkable polymorphism exhibited by these structures. Several protein sequences have been observed to adopt multiple distinct amyloid structures^{17,131,132}. The origins and implications of this polymorphism are a subject of debate, but it probably arises from the fact that these structures are not products of evolutionary selection. Consequently, numerous similar energy minima coexist within the amyloid energy folding landscape (Fig. 1a). A compelling argument supporting this notion is that functional amyloids, which have evolved towards the amyloid state, are largely monomorphic even under non-physiological conditions¹³³.

The ultrastructure of protein aggregates

The ultrastructure of protein aggregates varies considerably, ranging from cases involving the self-assembly of native-like folds¹³⁴ or 3D domain-swapped models¹³⁵ to primarily α -helical¹³⁶ or (most often) enriched in β -strands. The latter also form a spectrum of morphologies ranging from amorphous inclusions¹³⁷ to curly fibrils^{130,138} to highly ordered amyloid structures. What differentiates the highly ordered amyloid structure from less ordered forms of β -aggregation is unclear, given the latter still evades structural determination. Amyloid deposits in cells and tissues consist of long non-branched fibrous assemblies of thousands of peptides with a width of about 6–12 nm and lengths of up to several micrometres. Amyloid fibrils have a rope-like architecture composed of one or several protofilaments intertwined laterally along the fibril axis. The ultrastructure of amyloid fibrils often adopts a regular helical twist, but many also display a more planar ribbon-like structure. Successive advances in X-ray diffraction¹³⁹, crystallography⁴⁵, solid-state NMR^{140,141} and cryo electron microscopy (cryo-EM)¹⁴² have revolutionized structure acquisition and vastly improved resolution. Together these efforts have greatly advanced our understanding of the amyloid fold, resulting in an explosion in high-resolution structures¹⁹. These atomic resolution structures confirm that amyloids form their own class of protein assemblies sharing a distinct 'cross- β ' architecture^{139,143} that is not observed in other protein architectures^{4,5}.

We now have unprecedented detail on the structure of protofibrils and how they assemble into amyloid fibrils, thereby unveiling the structural origin of polymorphism. Importantly, structures of patient-derived fibrils have revealed the association of specific polymorphs to particular pathologies¹⁴². A β ₄₂ forms two distinct polymorphs associated with familial Alzheimer disease (FAD) and sporadic Alzheimer disease (sAD)¹⁴⁴ (Fig. 2), whereas tau forms ultrastructural polymorphs that are classified between different tauopathies and share several different protofilament folds^{145–147}. α S forms distinct fibrils in multiple system atrophy (MSA) and Parkinson disease^{148,149}. Notably, these structures are not representative of α S fibrils formed in vitro, which differ even further when incorporating disease mutations and post-translational modifications^{150,151}.

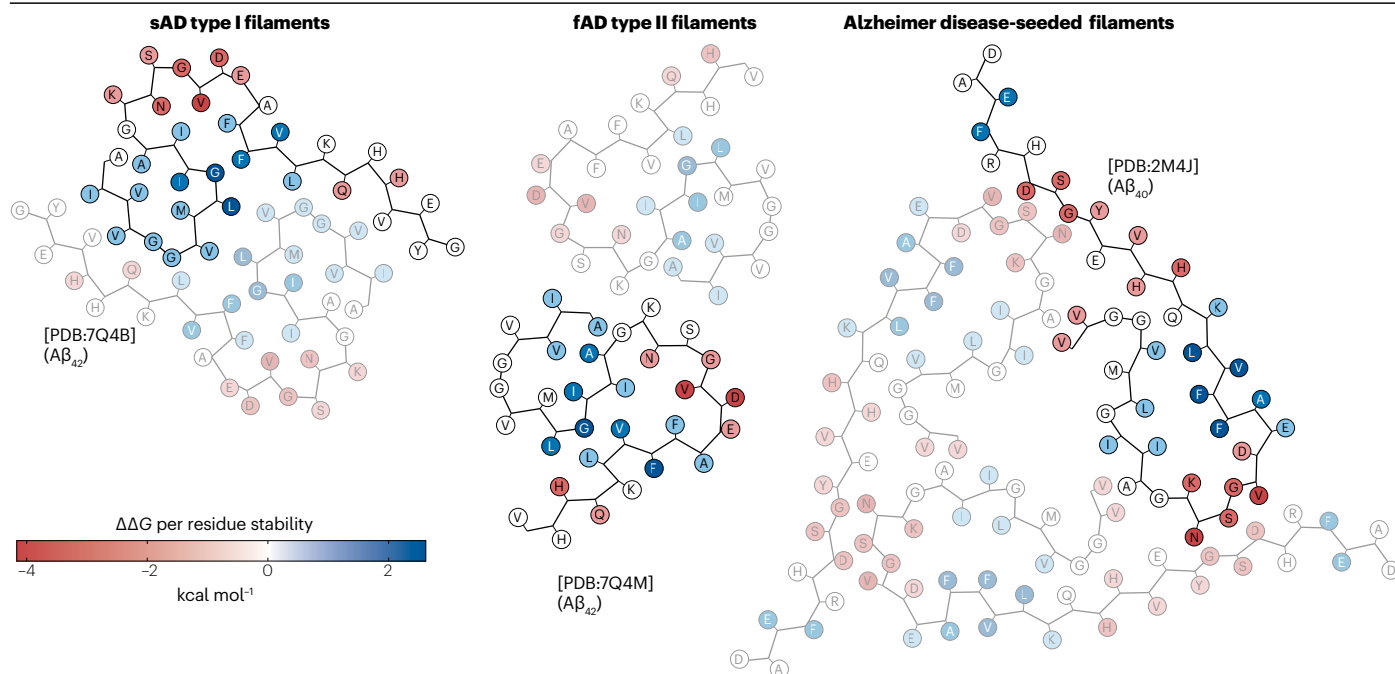


Fig. 2 | Structural determinants are shared between polymorphic amyloid fibril structures. Representations of one cross-sectional layer of polymorphic amyloid- β peptide ($A\beta$) fibril cores extracted from the brains of patients with different pathologies or post in vitro amplification (seeding). Structure (type I filaments) corresponding to the primary fold isolated from the brains of patients with sporadic Alzheimer disease (sAD) (left panel). Structure comprising two S-shaped packed protofilaments (type II filaments), mostly found in the brains of patients with familial Alzheimer disease (fAD) (middle panel). The latter was extracted also from several other pathologies, including patients with ageing-related tau astroglialopathy, Parkinson disease, dementia with Lewy bodies and frontotemporal dementia. Similarly, type I and type II filaments were recovered

in small amounts from fAD and sAD cases, respectively¹⁴⁴. Fibrils prepared in vitro after seeding $A\beta_{40}$ with extracts from patients with Alzheimer disease resulted in the formation of filaments with a distinct amyloid core (right panel). Individual residues are shown as circles and colour-coded based on the per residue energetic contribution to overall fibril stability. A single protofilament is highlighted in each case, together with transparent laterally packed identical protofilaments. Values were obtained from Stamp-DB, a public repository for the structural and thermodynamic classification of amyloid structures¹⁹. Despite their evident structural variability, different polymorphs share overlapping energetic frameworks composed of similar stabilizing sequence segments. PDB, Protein Data Bank.

Polymorphism has also been confirmed for TDP-43 (ref. ¹⁵²) and PrP¹⁵³, as well as in localized^{154,155} and systemic forms of amyloidosis, including ATTR amyloidosis¹⁵⁶, serum amyloid A (SAA) amyloidosis¹⁵⁷ and light chain amyloidosis¹⁵⁸. Conformational variants of various amyloids have been shown to stably self-propagate in cellular systems via the recruitment of soluble protein to the aggregates, and are characterized by diverse progression rates and distribution patterns^{159–162}. As a result, they are often referred to as prion-like strains and are invariably associated with different diseases, thus enabling accurate hierarchical classification of pathologies based on fibril structure¹³¹.

Although amyloid structure seems to be stable across patients sharing the same pathology, it is also clear that amyloids are extremely sensitive to environmental conditions. For example, distinct strains formed by $A\beta$ form at later stages of Down syndrome, suggesting that the convergence of ageing and disease-specific factors can dictate the formation of characteristic amyloid strains¹⁶³. Cryo-EM studies of in vitro-prepared amyloid fibrils of tau and αS have shown that additives such as heparin or nucleic acids can substantially alter amyloid structure^{164–166}, in a similar lateral mode of binding shown for amyloid-specific dyes¹⁶⁷. Faithful in vitro propagation of ex vivo polymorphs is challenging: although SAA patient-derived polymorphs seem to seed accurately in vitro¹⁶⁸, this is not the case for αS ^{169,170}. Finally,

in vitro assembly of tau polymorphs resembling disease-specific polymorphs is possible but highly dependent on buffer conditions, tau construct lengths and conditions of incubation, such as shaking^{171,172}. The contrast between seemingly conserved disease polymorphs in patients and the variability of in vitro amyloids suggests that the pathological conditions under which disease polymorphs emerge also reflect very specific pathophysiological contexts. Factors influencing polymorphic bias include disease-associated mutations¹⁶¹ and tissue or cell-specific proteolysis¹⁷³. Notably, post-translational modifications also have a significant role in this process. For instance, charge neutralization by acetylation of lysine chains in tau and phosphorylation (Tyr39) of αS favour β -stacking interactions^{150,174}. Additionally, ubiquitin incorporation has been proposed to alter protofilament packing in tau polymorphs in corticobasal degeneration (CBD) and Alzheimer disease¹⁷⁴. Patient-derived amyloid structures are often associated with undefined non-proteinaceous electron densities filling internal cavities or decorating surfaces of unknown origin, although they are suspected to be lipids or polyions of some sort¹⁴².

Structural determinants of amyloid stability

With hundreds of atomic resolution amyloid structures available, we can now identify general trends and properties determining amyloid

structure stability^{18,20}. One observation is that functional amyloids are generally associated with lower stability than disease-related fibrils (Box 1). Another immediate observation is that amyloid core structures do not always consist of full-length proteins^{171,175} but, rather, protein fragments. This observation suggests that large segments of protein sequences are not compatible with the amyloid state. Non-core segments constitute the fuzzy coat^{53,145} or are removed altogether by proteolysis. Removal of non-core segments seems to enhance amyloid stability, probably by exposing amyloidogenic fragments and removing incompatible ones¹⁷⁶.

Amyloid core structures are not uniformly stable upon inspection. Roughly one third of residues structurally support the cross- β conformation, whereas another third are frustrated due to unfavourable dihedral angles, main chain hydrogen bond geometries or suboptimal side chain packing; the final third contribute minimally to the amyloid fold. Structural frustration is mostly segmental and interspersed with segments forming regular β -strands. Thermodynamic analysis using empirical force fields such as Rosetta^{18,21} or FoldX²⁰ confirm these structural observations, indicating that structurally satisfied segments also form stabilizing tertiary side chain interactions within the amyloid protofibril. Comparing different polymorphs of tau, α S or A β shows that the same segments are structurally stabilizing in different protofibrillar folds forming very similar β -sheets. Not surprisingly, many of these correspond to APRs that have long been known to be crucial for amyloid assembly, such as PHF6 in tau (³⁰⁶VQIVYK³¹¹) or ¹⁶KLVFFA²¹ in A β (Fig. 3a–c; see Supplementary Table 1).

Protofibrillar polymorphs, therefore, share a common framework of cross- β -favouring segments that are stabilized by alternative tertiary side chain interactions. In the case of tau, stabilizing tertiary interactions are often provided by different framework segments that either pack against each other or at the same time also serve as anchor points providing stabilizing interactions to structurally variable segments, enabling these to be integrated in the cross- β core structure^{20,21,177} (Fig. 3d). The same modularity is also observed in other amyloids including A β or α S. Together these observations depict a highly frustrated and rugged polymorphic conformational landscape¹³². Unlike native protein folding, kinetic control of this energetic landscape has not been shaped by natural selection. Instead, the kinetics of amyloid assembly are directed by pathological interactions resulting in disease-specific polymorphic bias.

Kinetic principles of amyloid formation

Amyloid formation is a complex kinetic process that involves interconversions between aggregated species in an evolving reaction mixture that gradually enriches end-stage amyloids (Fig. 4). Several studies have determined the microscopic rates characterizing the key steps in amyloid assembly, including primary and secondary nucleation, elongation, fragmentation, oligomer disassembly and monomer dissociation. Nucleation, fibril elongation and fragmentation all contribute to the growth of mature amyloid fibrils and are counteracted by oligomer disassembly and monomer dissociation¹⁷⁸.

Nucleation

Primary nucleation (that is, the generation of new amyloid species in a pure solution of monomers) is entropically unfavourable, resulting in slow and stochastic growth rates¹⁷⁹. It involves the self-assembly of monomers into productive nuclei. Macroscopically, this kinetic step results in an initial lag phase representing the time required for the nuclei to form and grow to levels at which aggregation is detectable

through conventional methods (Fig. 4a). This process can be accelerated by heterogeneous interactions (for example, with lipids¹⁸⁰) that increase the local concentration of monomers and significantly speed up monomer conversion. Liquid–liquid phase separation has also been suggested to favour amyloid nucleation. Liquid–liquid phase separation involves the formation of liquid droplets, which again facilitates nucleation by increasing local protein concentrations.

Secondary nucleation is differentiated from primary nucleation by the fact that it occurs on the lateral surfaces of already formed fibrils and is catalysed by these surfaces. As lateral surface areas increase with fibril growth, secondary nucleation is thereby favoured, further accelerating aggregation. The difference in specificity between primary and secondary nucleation is still a matter of debate¹⁸¹. However, data from various studies suggest that secondary nucleation is a less specific process. For instance, proteome analysis of surface-induced co-aggregation in bodily fluids revealed that the composition of extracted deposits did not differ substantially. This finding was observed despite the bodily fluids being exposed to fibrils with diverse morphologies derived from different proteins¹⁸². Human prolactin and galanin, two proteins that co-localize in the same secretory granules yet are dissimilar in terms of the primary sequence, form functional co-assemblies through secondary nucleation, countering the notion that secondary nucleation is necessarily sequence-specific¹⁸³. Similarly, S100A9 co-aggregates by coating and reducing the availability of the same secondary nucleation surfaces that catalyse A β ₄₂ self-nucleation¹⁸⁴. Positive evidence supporting the specificity of secondary nucleation is mostly derived from research on A β assembly. For instance, evidence shows that surface catalysis is selective between A β ₄₀ and A β ₄₂ (ref. 185) as well as being enantio-specific¹⁸⁶. However, another study indicated that A β auto-catalysis occurs independently of surface properties, suggesting that the specificity of secondary nucleation for this peptide is driven by the intrinsic propensity of the free monomer to adopt a similar amyloid fold under given conditions¹⁸⁷.

Conversely, primary nucleation seems to be highly sensitive to sequence variation. Thermodynamic analysis of the elongation of aggregation cores through axial addition of monomers revealed that even a single residue variance accounts for >50% of the resulting sequences being structurally incompatible with cross-interactions¹⁸⁸. Primary A β nucleation is also sensitive to pH variation that alters net charges, whereas secondary nucleation is unaffected¹⁸⁹. Furthermore, intramolecular interactions play a crucial role during primary nucleation in facilitating the formation of initial nuclei by arranging monomers into a compact and nucleation-compatible conformation^{55–57}.

The morphological fidelity of secondary nucleation has also been debated. Structural evidence revealed that α S secondary nucleation is promoted by transient electrostatic interactions that involve the flexible ends of the protein excluded from the aggregation core and that these interactions promote the unfolding and local concentration of monomers, suggesting that the structure of the pre-formed core is not imprinted on the second-generation aggregates¹⁹⁰. In fact, amyloid strains formed through secondary nucleation have been suggested to be defined primarily by solution conditions rather than the structure of the templated fibrils^{187,191}. Independent studies for A β ¹⁹² and insulin fibrils¹⁹³ have also demonstrated that only the fibril ends faithfully template the morphology of derived aggregates. Cryo-EM studies have identified that under conditions conducive to templated structure formation, surface nucleation can still catalyse

different morphologies, indicating that secondary nucleation contributes to the increased variability in the formation of amyloid fibrils¹⁵⁵. However, observations of single A β fibrils using total internal reflection

fluorescence microscopy revealed that under the same conditions, secondary nucleation typically results in the formation of fibrils that resemble the parent fibril population¹³.

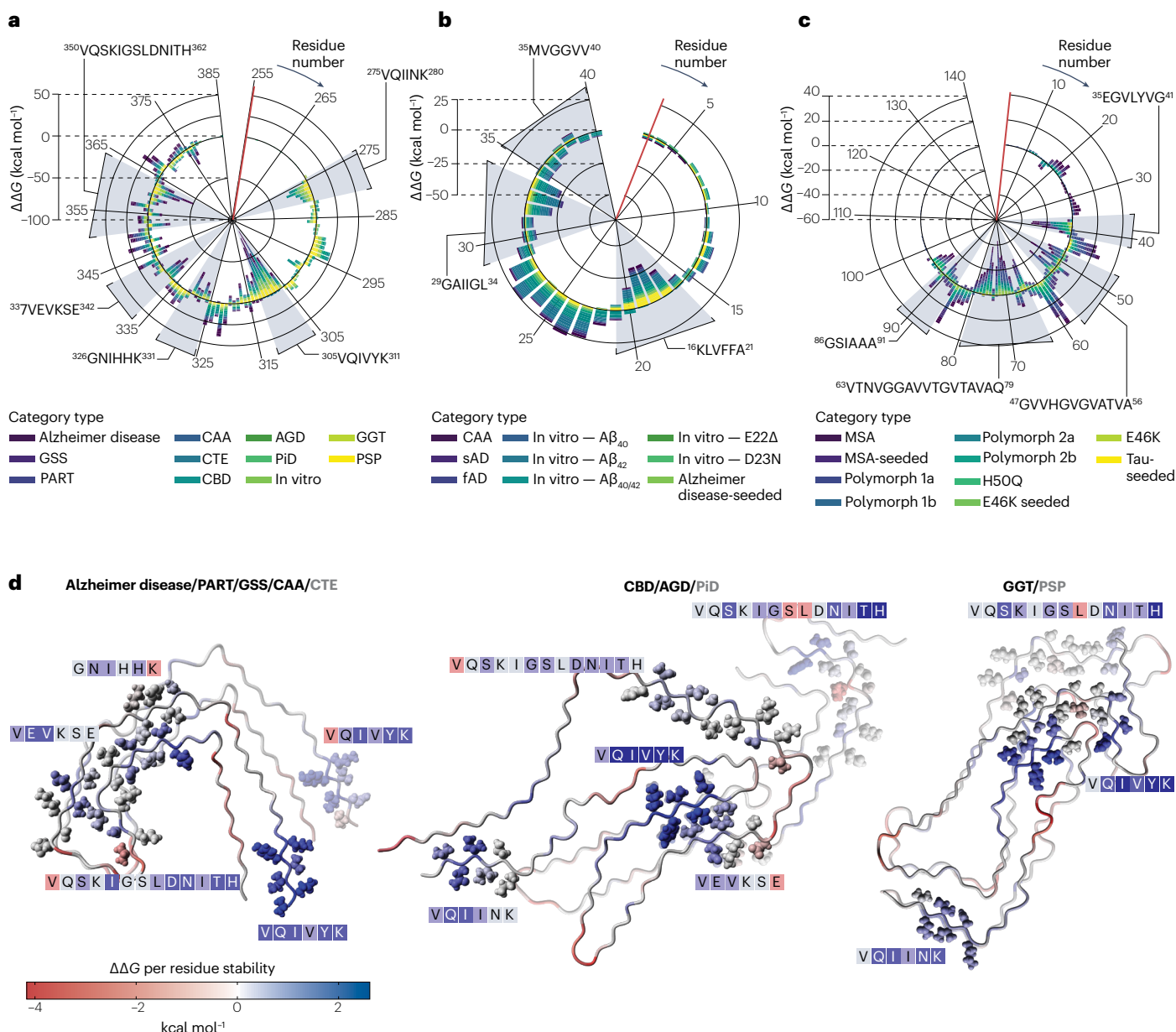


Fig. 3 | Structural determinants stabilize polymorphic amyloid fibril structures. **a–c**, Per residue cumulative energetic contributions (y axis) of polymorphic cores of amyloid fibrils formed by tau (panel **a**), amyloid- β peptide (A β) (panel **b**) and α -synuclein (α S) (panel **c**). Fibril structures used for the energetic calculations are categorized and colour-coded based on disease relevance for ex vivo structures and on disease mutations or experimental conditions for recombinantly produced fibrils, respectively (as shown in the figure key). Circular histograms, with protein residues shown along the perimeter, highlight that polymorphic cores share overlapping energy profiles with common stabilizing regions (shaded segments) that correspond primarily to previously experimentally determined aggregation-prone regions (APRs) of each protein^{47,160,177}, interspersed with regions of structural frustration. Circular histograms were generated using the ggplot2

library in R. **d**, A common framework contributes to the stability of tau fibrils derived from different pathologies. APRs form important stacking interactions that typically ‘clamp’ together the termini of fibril cores or participate in energetically pivotal packing interactions that have been identified by parallel thermodynamic studies^{19–21}. Cores are colour-coded based on per residue energetic contribution as described previously; chronic traumatic encephalopathy (CTE), Pick’s disease (PiD) and progressive supranuclear palsy (PSP) architectures are shown as overlaid transparent structures to highlight common interactions^{19,20}. AGD, argyrophilic grain disease; CAA, cerebral amyloid angiopathy; CBD, corticobasal degeneration; fAD, familial Alzheimer disease; GGT, globular glial tauopathy; GSS, Gerstmann–Straussler–Scheinker disease; MSA, multiple system atrophy; PART, primary age-related tauopathy; SAD, sporadic Alzheimer disease.

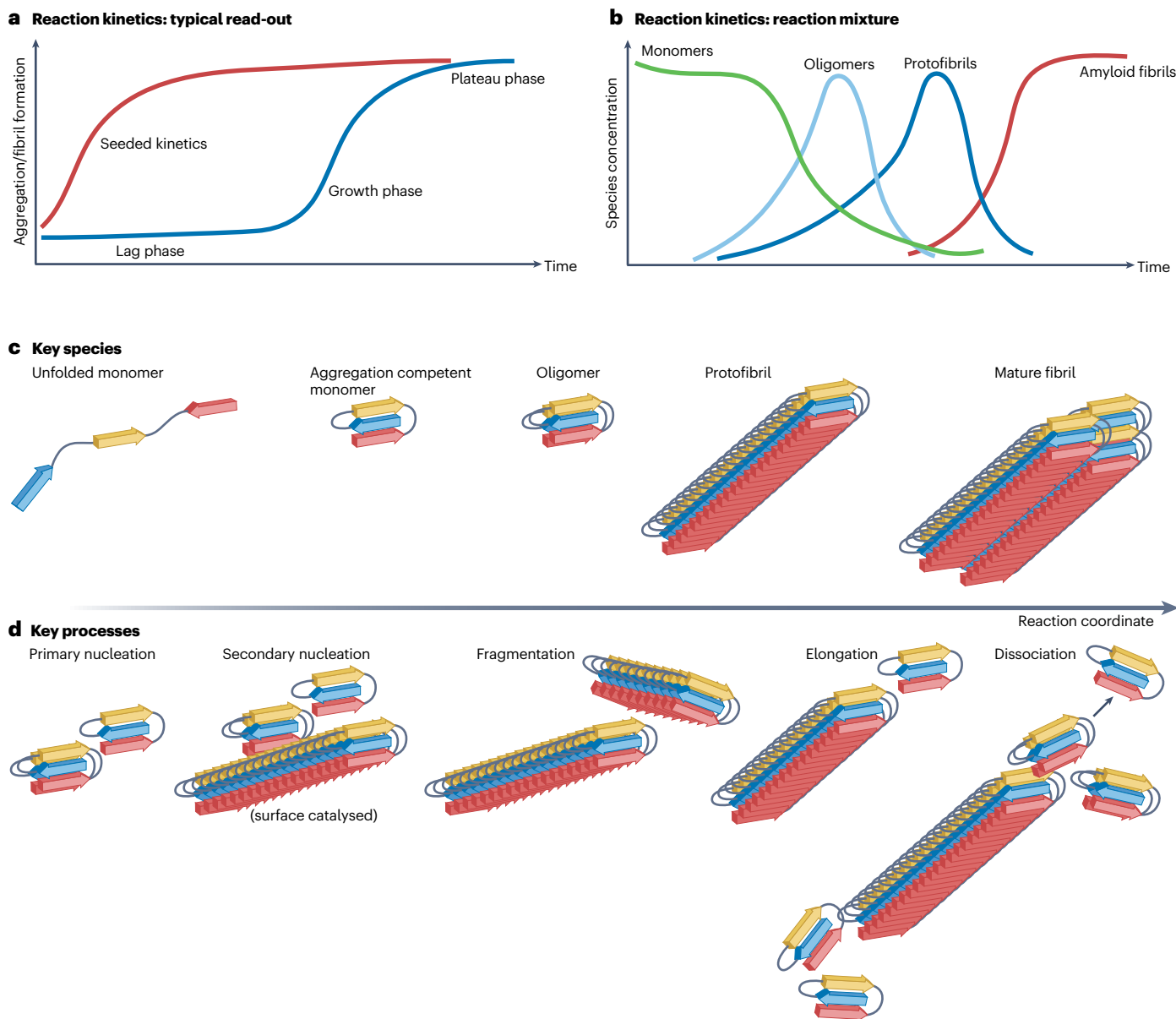


Fig. 4 | Overview of protein aggregation kinetics and processes that contribute during amyloid fibril formation. **a**, Mainstay kinetics of amyloid fibril formation (blue line). Kinetics are characterized by a starting lag phase during which initial nuclei and ‘on-pathway’ oligomers are formed and reach aggregate concentrations that can be monitored in bulk aggregation assays. This lengthy lag phase can be surpassed by seeded experiments (shown by red line) during which pre-formed aggregation nuclei are supplied exogenously. This process is followed by a shorter growth phase during which the rapid recruitment of monomers is catalysed by fibril ends or by secondary nucleation on lateral surfaces that overtakes the aggregation process. Eventually, the process reaches an equilibrium that appears as a plateau phase in typical read-outs. **b**, Overlapping presence of different species during different phases of aggregation growth. During early phases of nucleation, it has been suggested that monomers adopt a compact aggregation conformation that can gradually lead towards the formation of oligomeric and prefibrillar species. During elongation, these structures can

promote the growth of protofilaments and mature fibril aggregates that are formed through lateral interactions of the former. **c**, Key species during the formation of amyloids. **d**, Key kinetic processes that take place during the formation of amyloid aggregates. Templatting that proceeds through primary nucleation results in slow and stochastic growth rates¹⁷⁹, whereas secondary nucleation becomes proportionally faster based on amyloid growth, as it occurs on the surfaces of pre-formed aggregate species. Aggregate growth is then promoted by further incorporation of monomers during elongation, whereas fragmentation, which can occur throughout the kinetic process, does not change the number of aggregates but can shift the balance between processes (for instance, primary and secondary nucleation) as it alters the number of reactive sites by modifying the numbers of reactive species. The dissociation of oligomers does not directly reduce aggregation growth but can do so indirectly by replenishing the monomer pool. Similar monomer dissociation can also occur directly from fibril tips.

Elongation

Single-molecule observations have shown that fibril elongation for various amyloidogenic proteins (both functional and pathological) occurs through a stop-and-go mechanism^{194,195}. This process is defined by growth bursts (of constant rates) with intermittent stationary periods¹⁹⁶. Importantly, amyloid elongation might be polar, suggesting that fibril tips are not equally reactive in the recruitment of monomeric proteins to the growing amyloid fibril^{195,197}. This unequal reactivity has also been attributed to imperfections occurring at fibril tips, reminiscent of a 'lock-dock' mechanism for elongation¹⁹⁸ in which the addition of monomers at fibril tips requires significant structural rearrangement of the newly added monomer¹⁹⁹ as well as of the molecules directly in contact at the fibril tip²⁰⁰. Monomer dissociation from fibril tips has also been observed to occur at rates that supersede incorporation of monomers; however, this balance is dependent on the relative monomer concentration²⁰¹.

Fragmentation and dissociation

Fragmentation is strongly dependent on environmental conditions, such as temperature and mechanical perturbations²⁰², but also relies on the intrinsic propensities of protein sequences and the morphological properties of the fibrils²⁰³. Fibril stability and length distributions are important determinants of fibrillar brittleness²⁰⁴. Fragmentation has also been linked to amyloid toxicity, as it influences cellular uptake of amyloid fragments generated through mechanical fragmentation²⁰⁵. This process is associated with the average size of aggregates, which exhibits an inverse correlation with the cellular uptake of amyloids²⁰⁶. Fragmentation is also associated with the generation of reverse oligomeric species and is suppressed when a critical fibril mass is reached²⁰⁷. These species are usually 'on-pathway' as they resemble the parent fibril structures and can deviate in terms of function, in contrast to oligomers formed directly from monomer nucleation. Reverse-generated fibrils derived from mechanical fragmentation of A β amyloid fibrils interact with a central APR of the monomeric form similar to end-state fibrils, whereas oligomers formed during the nucleation process preferentially interact with the C-terminal end²⁰⁸. On the other hand, oligomers formed during the nucleation of monomers are largely non-fibrillar and predominantly dissociate to replenish the monomeric pool instead of converting to mature fibrils^{209,210}. The formation of such off-pathway oligomeric species is proposed to compromise the fidelity of aggregation kinetics by increasing the variability of lag phases and the amplitude of end-state phases²¹¹. Early metastable oligomers inhibit nucleation and growth of fibrils by competing for monomers, thus generating biphasic assembly kinetics that transition to standard sigmoidal growth over a critical monomer concentration that hinders their formation²¹². Furthermore, these metastable species block secondary nucleation by attaching to fibril surfaces²¹³. Off-pathway oligomers have been linked to higher levels of disorder that create repulsions or introduce entropic costs for the formation of ordered fibrils²¹⁴.

Interactions and toxicity of amyloids

The mechanisms through which amyloids exert toxicity have been a topic of ongoing debate in the field. Multiple lines of evidence suggest that small soluble species are responsible for the observed toxicity. These species have been associated with numerous detrimental effects, such as permeabilization of cellular membranes, impairment of degradation pathways, disruption of synaptic signalling and mitochondrial dysfunction^{215–217}. The toxicity induced by fibrillar aggregation

is attributed to mechanical perturbations, sequestration of cellular factors and the subsequent activation of inflammatory responses²¹⁸. Both lateral and axial surfaces of amyloids can engage in interactions with other biomolecules, including charged molecules such as polyphosphates and polysaccharides, as well as with lipids and other proteins (Fig. 5a–c).

Although the precise identity of many toxic species and their interfaces remains to be fully elucidated, the importance of heterotypic interactions in shaping the properties and infectivity of amyloids or promoting their toxic phenotype (see next section) is now firmly established. A prominent example includes the cross-interaction between medin and A β , which initiates the vascular deposition of the latter²¹⁹. Synergistic interactions between different disease-associated proteins (including tau, α S and A β) typically exacerbate neuropathology^{220,221}. For instance, co-aggregation of β_2 -microglobulin with A β promotes cognitive decline in Alzheimer disease²²². Tau aggregates originating from PS19 mice provide a compelling illustration of how heterotypic interactions can exert toxicity through various pathways. In reporter cells, PS19-derived tau seeds impair chaperone activity²²³, whereas in mouse models of tauopathy they inactivate histone demethylase LSD1 through sequestration (thus linking tau aggregation to downstream neuro-dysfunction pathways)²²⁴. A study of α S pathological polymorphs derived from either glial cytoplasmic deposits or Lewy bodies revealed that they are shaped by intracellular milieus and, in turn, associate with different toxic effects²²⁵. Finally, a little considered aspect is that amyloid fibril surfaces can have catalytic properties^{85,226,227} (Box 4), which might have biological consequences; for example, fibrils of A β catalyse the hydrolysis of neurotransmitters, such as dopamine and adrenaline²²⁸.

Amyloids engage in diverse interactions

The fibril tips of amyloids are formed by partially buried hydrophobic residues of the top rungs¹⁸. Perhaps unsurprisingly, the growing ends of amyloids have been identified as the primary cross-interface with lipid vesicles^{229,230}. A comprehensive analysis has revealed similar findings for heterotypic protein interactions¹⁸⁸ (Fig. 5c). In particular, the findings suggest that sequence similarity plays a crucial role for the cross-reactivity of amyloid fibril tips. Registered stacking of identical side chain residues on top of each other along the fibrillar axis enables sequence-dependent templated binding, resembling the self-interactions formed during fibril elongation¹⁸⁸. Importantly, design strategies employing these properties have been used in several studies to develop novel therapeutics that target amyloids through structure-based inhibitor designs^{154,188,231–234}. In these designs, the inhibitor peptides are constructed based on the APR sequence to enable binding to the fibril tip. However, these inhibitors also incorporate sequence variants in one or a few positions that block further fibril growth. Inhibitors initially developed to target APRs of A β assembly were found to also inhibit islet amyloid peptide (IAPP) growth due to the sequence similarity of their respective APR sequences²³⁵. This observation underscores the sequence specificity of APR cross-interactions but also their ability to tolerate mismatches. Similar rules of specificity seem to support amyloid co-assemblies. Solved structures of heterocomplexes such as A β_{40} with A β_{42} or the necrosome complex (which is a functional amyloid hetero-assembly composed of successive rungs of RIPK1 and RIPK3; see Box 1 and Supplementary Table 1) have validated the necessity for sequence and structural compatibility along the fibril axis^{236,237}. It is worth noting that such interactions actively steer amyloid polymorphism by altering the morphology of fibrils formed by self-assembly^{188,208}.

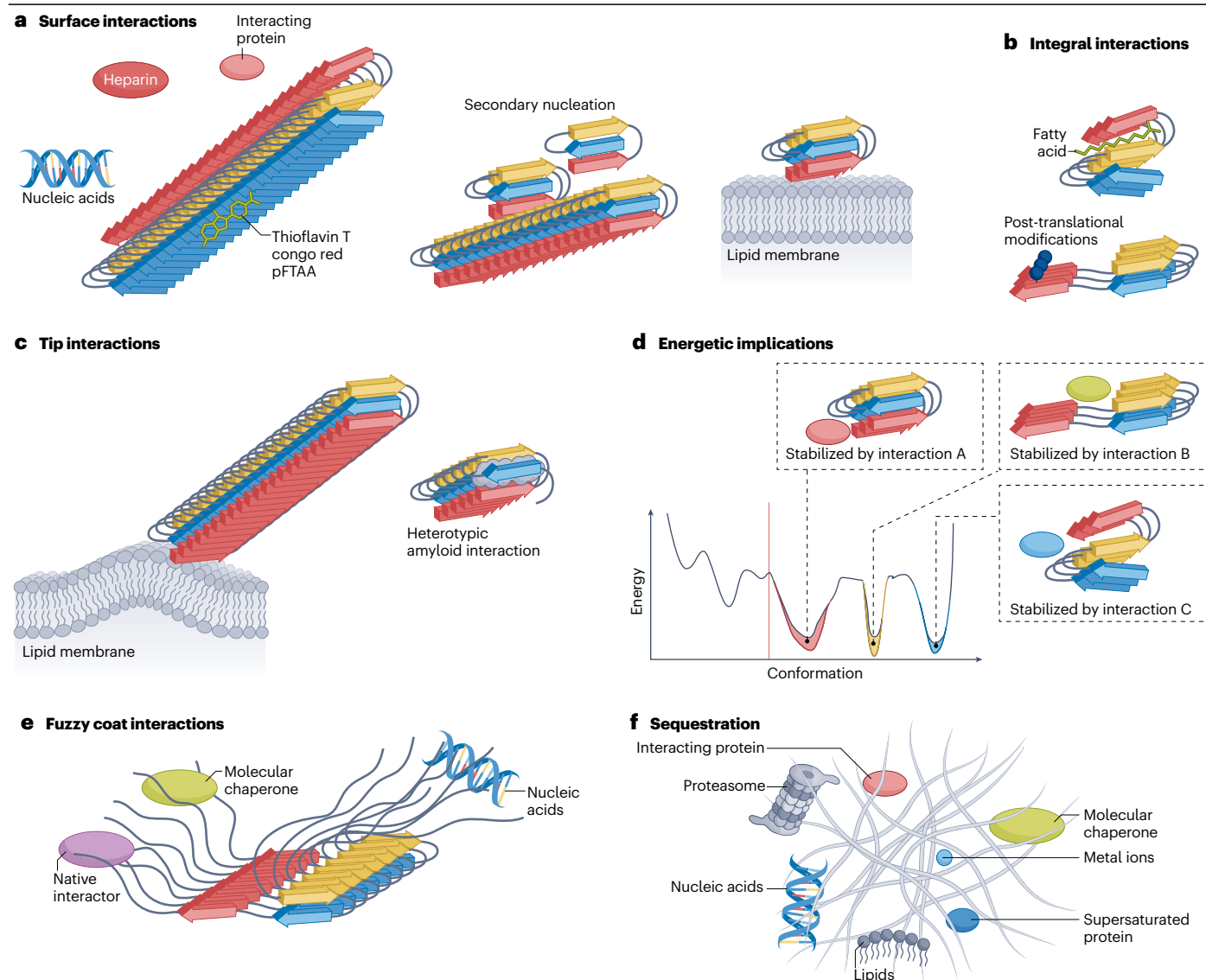


Fig. 5 | Amyloid interactions with other biomolecules can have significant implications for aggregation-related mechanisms of toxicity. **a**, Lateral surfaces of amyloid fibrils have been proposed to self-catalyse their growth through secondary nucleation, but also to facilitate cross-interactions with important cofactors such as polyanionic nucleic acids and polysaccharides. The same surface might also promote interaction with lipid bilayers. **b**, Heterotypic interactions with other molecules (such as lipids) or integral sequence properties (such as post-translational modifications) can assist with shaping the morphology of derived aggregates. **c**, The growing ends of amyloid fibrils are reactive surfaces that promote heterotypic interactions with other proteins or lipid bilayers. **d**, Heterotypic interactions with other cofactors can facilitate stabilization and

steer the morphology of amyloid fibril structures by lowering the energy barriers of specific polymorphs. **e**, Non-core components of fibrils, such as the fuzzy coat, have also been shown to promote the functionality of fibrils by harbouring interactions with other biomolecules⁵³. **f**, The heterogeneity of amyloid deposits in patients can be explained by several mechanisms of toxicity. The interplay of amyloids with the proteostatic machinery, including molecular chaperones, as well as the presence of a metastable sub-proteome that exists beyond its solubility limit and is therefore susceptible to widespread proteostatic collapse could explain the heterogeneous composition of in vivo deposits. Natural cofactors of fibrils are often co-deposited with amyloids and could lead to both a loss-of-function phenotype or gain-of-function toxicity events.

Owing to identical side chain stacking, the lateral surfaces of amyloid fibrils form regular steric patterns along the axis that are either hydrophobic or polar/ionic in nature, thus enabling repeated binding of diverse molecules along the fibril axis (for example with amyloid reporting dyes^{84,238}) (Fig. 5a). As a consequence of this lateral periodicity, unidentified electron densities are often found decorating the

perimeter of amyloid structures (defined by cryo-EM), suggesting the presence of repetitively bound cofactors²³⁹. These densities are often adjacent to stacks of same-charge side chains, suggesting the binding of charge-compensating polyionic cofactors^{239,240}. Lateral surface templating has been associated with co-aggregation events involving several different proteins that do not share sequence or structural

architecture^{184,213,232,241}. Cofactor binding therefore likely contributes to steering amyloid polymorphism in different pathological contexts (Fig. 5d).

The dynamic and structurally disordered regions surrounding the rigid cores of amyloids are referred to as fuzzy coat regions. As a result of their increased surface accessibility and structural dynamics, these sequences are believed to modulate various amyloid activities, such as spreading, liquid–liquid phase transition and cross-interplay with other cellular components^{53,242} (Fig. 5e); for example, the functional oligomerization of Orb2 enhances translation²⁴³. Similarly, these regions can modulate amyloid assembly by mediating chaperone binding²⁴⁴, or can lead to loss-of-function events by dragging native interactors towards co-aggregation²⁴⁵.

Sequestration of proteins

Protein aggregates can sequester cellular components, causing loss of function or mis-localization (Fig. 5f). For example, TDP-43 fragments form cytoplasmic inclusions that sequester RNA-binding proteins, leading to dysregulation of mRNA maturation²⁴⁶. FUS mis-localization in the cytosol alters its RNA regulatory functions and induces DNA damage²⁴⁷. Dipeptide repeat expansions resulting from C9orf72 hexanucleotide repeat expansions, which pathologically assemble in frontotemporal dementia and amyotrophic lateral sclerosis (see Supplementary Table 1), deplete levels of functional pATM and hnRNPA3, causing breaks

in double-stranded DNA²⁴⁸. Mis-localization of the nuclear histone demethylase LSD1 to cytoplasmic neurofibrillary tangle tau inclusions in P301S mouse models promotes neuronal cell death in the hippocampus and cortex leading to associated learning and memory defects²²⁴. Similarly, APRs can serve as heterotypic anchors, facilitating co-assembly of functional proteins and increasing cellular susceptibility to aggregation spreading (leading to loss-of-function toxicity or to a gain-of-function phenotype)^{188,208}. Several studies have shown that the heterotypic composition of condensates can synergistically trap proteins towards co-aggregation following their maturation^{220,249}. Notably, however, contradictory theories propose that hetero-interactions can also work as a buffer to control protein aggregation within condensates²⁵⁰.

Sequestration can also disrupt cellular proteostasis. PolyQ expansions from huntingtin or ataxin 7 sequester the HSP70 chaperone, leading to proteostasis imbalance and aggregation of ataxin 3 (ref. 251). Tau extracts from the brains of PS19 mice impair protein folding and clathrin-mediated endocytosis by inactivating HSP70, HSP90 and J-domain chaperones, an effect that can be reversed by exogenous addition of a small molecule that replenishes cytosolic chaperone expression²²³. RNA-induced tau and A β aggregates co-deposit ribosomal 80S, which reduces translation efficiency in yeast cells²⁵². Cryo electron tomography showed that intracellular inclusions formed by different aggregation-prone polypeptides recruit and impair the functionality of proteasome subunits (Fig. 5f), along with TRiC/CCT

Box 4

Synthetic amyloids

Increasingly, synthetic amyloid fibrils are being explored for various applications, ranging from structural scaffolds to improved techno-functional properties in food, innovative catalysts³⁸⁵ and the deliberate knockdown of target proteins by targeted aggregation.

Materials

The chemical stability, fibrillar nature and tensile strength of amyloid fibrils makes them attractive components of novel materials³⁸⁶. Moreover, amyloid fibrils can be generated from short peptides whose sequence can be rationally designed. For example, synthetic spider silk was constructed on the basis of amyloid architecture and showed tensile strength and toughness that surpassed some naturally occurring silks³⁸⁷.

Catalysis

Amyloids initially appeared in catalytic applications by tethering existing enzymes to an amyloid backbone^{388,389}, but increasingly the surface of the amyloid itself is being employed directly for catalysis, either through the immobilization of metal ions^{85,226,227}, prosthetic groups or co-enzymes^{390–392}, or simply by the positioning of charged side chains on the amyloid surface³⁹³. The concept of surface-induced catalysis suggests that coordinated interaction interfaces are essential for this functional process. However, the cryo electron microscopy structure of a catalytic amyloid revealed that these too are susceptible to polymorphism³⁹⁴; specifically, metal-binding sites were observed to form at the interface of

protofilament contacts, indicating that polymorphism was instead critical in enabling the functionality of these filaments.

Food

Proteins have a techno-functional role in food applications, such as gelling, foaming and emulsifying. Evidence indicates that the fibrillar state of food-born proteins can improve (some of) these properties and that common food processing techniques promote their formation^{310–315}. Given that bioavailability is higher in the amyloid form (in which only the amyloid core is protected from proteases) than for the folded protein, increasing the fibrillar protein content of certain foods might one day lead to a reduced need for protein, in particular of animal-derived protein^{138,395–397}, although the impact on human health requires further investigation³⁹⁸.

Targeted aggregation

Our group has shown that short aggregation-prone peptides can be used to induce the aggregation of selected target proteins. This approach is based on the observation that most naturally occurring polypeptides contain aggregation-prone regions (APRs) that tend to be unique within their proteome, and that amyloid assembly displays sequence specificity³⁹⁹. We have shown the potential of this approach in combating bacterial strains that are resistant to established antibiotics^{70,400,401}, in viral infections⁶⁹ and in an oncological setting^{68,71}. Moreover, derivatives labelled with radionuclides have potential as PET tracers⁴⁰².

and ribosomes, providing visible intracellular evidence of spatial isolation of proteostatic cellular components by protein aggregates^{253,254}. Importantly, however, proteasome inactivation has also been proposed to occur in a non-sequestered fashion. Oligomeric species derived from α S, A β and Htt-53Q sharing significant structural similarities bind with high affinity and inhibit the functionality of the 20S proteasome through allosteric impairment²⁵⁵. Cryo electron tomography also revealed that amyloid-like fibril aggregates can cause gain-of-function toxicity resulting in lysosomal defects²⁵⁶ and impair vesicular trafficking as shown in cases of Alzheimer disease and Parkinson disease pathology²⁵⁷.

Sequestration by amyloids can also be polymorph-specific. For instance, single-particle analysis showed that a size-dependent population of patient-derived aggregates exerts its toxicity by penetrating cells and binding to proteasome subunits, causing their inactivation²⁵⁸. Differential proteasome inhibition was also observed for two distinct α S strains prepared under different in vitro conditions²⁵⁹.

Membranes and lipids

Lipid-amyloid interactions have long been recognized as a source of toxicity in proteinopathies. For instance, Lewy body formation, involving the incorporation of membranous organelles, has been proposed as a major contributor to toxicity in Parkinson disease, surpassing that of α S fibrillation²⁶⁰. However, the exact impact of lipid interactions on the morphology and toxicity of aggregates is still a subject of debate. The functionality of α S in presynaptic vesicle docking is affected by the composition of biological membranes²⁶¹. In line with this observation, pathological aggregation of α S has been proposed as an after-effect of its dysregulated binding to its natural substrate PIP3, a process that is initiated by a loss of function of synaptojanin 1, a phosphoinositide phosphatase and a risk gene for early-onset Parkinson disease²⁶². Similarly, apolipoproteins participate in native-like lipid interactions in a mechanism evolved to protect against the exposure of their APRs and their conformational switching from α -helical to cross- β amyloid-compatible folds^{263,264}. NMR spectroscopy identified that early APR-driven interactions enable the partial insertion of A β into membranes, a process that promotes nucleation and shapes the morphology of the derived fibrils²⁶⁵. Similar interactions promote the segmental assembly of α S on the surface of anionic phospholipids²⁶⁶. Free lipids are proposed to shield the hydrophobicity of exposed APRs in a chaperone-like manner, but, above a critical concentration, they can also enable the insertion of oligomeric species into bilayers⁵⁹.

The molecular composition of membranes also determines their differential binding affinity to different amyloid species. For example, lipid bilayers generated by lipid oxidation either promote the nucleation of monomeric A β , stabilize non-toxic oligomers that are still capable of compromising the integrity of bilayers or reform on-pathway protofibrils that exhibit clear toxicity²⁶⁷. Similarly, ApoE4, a genetic risk factor for Alzheimer disease, is linked to lipid dysregulation in Alzheimer disease brains; it impairs cholesterol transport causing intracellular lipid deposition²⁶⁸ and myelination dysfunction²⁶⁹. However, ApoE4 also accelerates A β deposition²⁷⁰ and hinders its clearance²⁷¹; therefore, its exact role in Alzheimer disease remains unclear.

Lipid-oligomer interactions are in the spotlight for understanding amyloid toxicity. Cryo-EM determination of oligomers formed by A β when fused to an α -haemolysin toxin revealed that it forms a lipid-soluble heptameric pore complex²⁷², reminiscent of the formation of similar pores by other amyloid-forming proteins^{273,274}.

Lipid-oligomer interactions have also been labelled as highly dynamic in several systems. Solution and solid-state NMR have revealed that two different species of α S oligomers are largely disordered or require only local folded elements when disrupting membrane integrity²⁷⁵. Soluble oligomeric A β species formed at different maturation stages induce membrane permeabilization or an inflammatory response as a function of size²⁷⁶. Conversion of dynamic non-amyloid-like oligomeric IAPP species has been linked to gain-of-function toxicity by inducing vesicle perforation²⁷⁷. However, another study showed that IAPP-induced vesicle leakage is a biphasic process mainly attributed to secondary nucleation and the gradual elongation of fibrillar seeds in the presence of membrane particles of various lipid compositions²⁷⁸.

Mature amyloid fibrils have been primarily linked to toxicity as inducers of mechanical stress that can cause tissue damage, which partially also includes damaged membrane components²⁷⁹. Direct structural evidence confirms the binding of lipids to mature amyloid fibril aggregates²³⁰. Cryo-EM revealed that α S forms distinct fibrils in the presence of lipids, which in turn were found to decorate their exposed hydrophobic patches, liaise as cross-contacts for their alternative quaternary organization or channel through internal cavities of the amyloid core²⁸⁰. Correlative light and electron microscopy analysis of Lewy body inclusions from Parkinson disease post-mortem human brain tissue exposed a crowded presence of lipid components²⁸¹. Similar findings were revealed by cryo electron tomography in deposits formed in a cell culture reporter model of systemic AA amyloidosis²²⁹. Impressively, this study also identified that the primary site of interaction between amyloids and lipid vesicles corresponded to the fibril tips rather than their lateral surfaces, suggesting an amyloid-driven specificity for lipid interactions²²⁹ (Fig. 5c). This finding might explain why evidence of lipid interactions is scarce despite the numerous solved ex vivo structures. Stringent purification protocols for aggregate extraction might promote this selective bias, although analysis of lateral amyloid surfaces indicates that fibril cavities are typically narrow and occupy polar residues, thus hindering lipid binding⁸⁴. Indeed, efforts tracing aggregate surface hydrophobicity of α S aggregates revealed that oligomeric species expose more hydrophobic patches than their fibril counterparts²⁸². Confirming the above, sub-micellar lipids were shown to catalyse apolipoprotein C-II aggregation, yet no lipids were recovered as cofactors in contact with mature fibrils. Similarly, cholesterol-containing lipid surfaces seemed to massively catalyse primary heterotypic nucleation of A β ²⁸³, whereas aminosterol derivatives can facilitate the secondary nucleation of A β and reduce its toxicity in *Caenorhabditis elegans* either by competing for surfaces (and thus causing the displacement of oligomers) or by promoting the formation of less toxic mature species²⁸⁴.

The emerging role of microbiomes

External factors have also been associated with aggregation disorders although the exact nature of this relationship is yet to be defined. Pathogen-associated molecular patterns (PAMPs; short conserved microbial motifs that are absent in the host) from bacterial infections activate immune responses, triggering microglia activation and oxidative stress (which are both downstream responses linked to neurodegeneration). Accumulation of endotoxin lipopolysaccharide (LPS)²⁸⁵ owing to systemic infection is associated with various aggregation-based pathologies²⁸⁶, inducing neuroinflammation²⁸⁷ and morphological changes in amyloid fibrils²⁸⁸. Interestingly, the amyloidogenic SAA protein suppresses the effects of LPS and ameliorates toxicity in mice²⁸⁹.

The gut–brain axis has a role in aggregation diseases, but this topic is still in its infancy. Both the composition of the gut microbiome and dietary changes affect Alzheimer disease and Parkinson disease^{290,291} to the level that dietary restructuring is considered a therapeutic strategy against disease progression^{292,293}. The composition of the gut microbiome has also been linked to levels of disease biomarkers such as ApoE²⁹⁴. Microbiome-generated functional amyloids can interact with brain amyloids, influencing aggregation and causing behavioural and motor impairments^{295–297}. Bacterial amyloid chaperones can inhibit pathological amyloids²⁹⁸, whereas amyloids themselves can exhibit antimicrobial activity^{299,300}. For instance, A β can form toxic oligomeric pores with bactericidal effects³⁰¹; sequence similarity between A β and bacteriocins might have evolved as a strategy to tune an immune response against the latter³⁰¹.

The interplay between the intestinal environment and amyloids has also been exposed. For instance, intestinal defensins with antimicrobial activity cross-interact and prevent further propagation and cell cytotoxicity of A β , IAPP and calcitonin³⁰². Microbiota-excreted fatty acids enhance A β plaque formation in mouse models of Alzheimer disease³⁰³. In light of the ability of other metabolites to template amyloid formation^{304,305}, this concept raises further questions regarding the role of diet and food in amyloid pathologies^{306,307}. Indeed, food proteins have been shown to form amyloid fibrils in isolation (Box 4), as well as to cross-seed, resulting in inhibition³⁰⁸ or exacerbated pathological amyloid formation³⁰⁹. These findings raise concerns in terms of the safety and impact on health of common food processing approaches that rely on aggregate formation, such as gelation or foaming^{310–315}.

Finally, the relationship of amyloids to viruses is also gaining traction. For example, the HSV-1 viral corona promotes aggregation of A β both in vitro and in animals³¹⁶. Similar cross-interactions have been validated between SARS-CoV-2 (and other viral infectious agents) and amyloid proteins³¹⁷. The SARS-Cov-2 nucleocapsid protein utilizes electrostatic interactions to accelerate α S fibrillation and induce cell death³¹⁸. In fact, parallel studies have shown that SARS-Cov-2 proteins can also form amyloids in isolation^{319,320}, with similar results deriving for other viral species, such as the Nipah and Hendra viruses³²¹. Viral infectivity has also been indirectly associated with neuroinflammation, protein quality control dysfunction and spreading of misfolded products^{322,323}; protein aggregation has been proposed as a strategy for the development of novel types of antiviral agents⁶⁹.

Conclusions and perspectives

Progress in treating protein misfolding diseases is evident from US Food and Drug Administration (FDA) and European Medicines Agency (EMA) approval of therapies such as tafamidis for patients with familial amyloid polyneuropathy³²⁴, or migalastat, which rescues the folding of α -galactosidase in Fabry disease. Voxelotor has shown promise as a conformational corrector of haemoglobin in sickle cell disease³²⁵, similar to several approved correctors of the folding of the CFTR conductance channel (ivacaftor, tezacaftor and elexacaftor). Even in the challenging field of neurodegenerative diseases, clinical trials with lecanemab in Alzheimer disease have shown promising progress. Specifically, selected patients with early mild cognitive impairment, even at later stages of the disease, experienced a moderate slowdown in the decline of cognition and function³²⁶. This breakthrough comes after years of uncertainty regarding the role of aggregates in Alzheimer disease, compounded by unsuccessful clinical trials, controversial studies and a lack of effective treatments^{327,328}. Our deepened understanding of the structural characteristics, kinetics and interactions of amyloid

structures has led to significant advancements in the development of engineered therapeutics. Notably, there has been an increase in structure-based approaches focused on the creation of highly potent and specific anti-amyloid biomimetics^{296,329,330}. These designs aim to target the attachment to growing fibril ends and thus block axial propagation, effectively combatting the assembly of various proteins such as tau^{188,231}, α S²³², A β ²³³, IAPP¹⁵⁴ and transthyretin (TTR)²³⁴. Heterotypic recognition has also been employed to raise antibodies^{331,332} and to engineer custom protein designs²³² that slow down the formation and toxicity of aggregates. Similarly, our improved understanding of the factors dictating amyloid formation and stability is now being harnessed to develop synthetic functional biomaterials with diverse applications, including catalysis, supporting scaffolds and bioproducts in the food industry (Box 4).

These encouraging results, along with breakthroughs in protein structure prediction from artificial intelligence-based approaches (Box 3) and direct structural determination of amyloids from patient tissues, are galvanizing the field and laying the groundwork for future breakthroughs. Despite this progress, our understanding of protein aggregation is still in its infancy. Critical features, such as the spatiotemporal dynamics of early amyloid formation, the diversity of oligomers, the involvement of on and off-pathway species, and the true origins of toxicity in disease, remain subjects of debate. It is increasingly evident that multiple overlapping processes and pathways contribute to protein aggregation. However, we are now reaching a point at which we can deduce the structure–function relationship of these species more accurately. These insights provide hope that the fundamental research discussed in this Review will ultimately benefit patients worldwide, improving their quality of life.

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References

- Hipp, M. S., Kasturi, P. & Hartl, F. U. The proteostasis network and its decline in ageing. *Nat. Rev. Mol. Cell Biol.* **20**, 421–435 (2019).
- Stroo, E., Koopman, M., Nollen, E. A. A. & Mata-Cabana, A. Cellular regulation of amyloid formation in aging and disease. *Front. Neurosci.* <https://doi.org/10.3389/fnins.2017.00064> (2017).
- Hartl, F. U. Protein misfolding diseases. *Annu. Rev. Biochem.* **86**, 21–26 (2017).
- Chiti, F. & Dobson, C. M. Protein misfolding, amyloid formation, and human disease: a summary of progress over the last decade. *Annu. Rev. Biochem.* **86**, 27–68 (2017).
- Buxbaum, J. N. et al. Amyloid nomenclature 2022: update, novel proteins, and recommendations by the International Society of Amyloidosis (ISA) Nomenclature Committee. *Amyloid* **29**, 213–219 (2022).
- Wechalekar, A. D., Gillmore, J. D. & Hawkins, P. N. Systemic amyloidosis. *Lancet* **387**, 2641–2654 (2016).
- Jucker, M. & Walker, L. C. Propagation and spread of pathogenic protein assemblies in neurodegenerative diseases. *Nat. Neurosci.* **21**, 1341–1349 (2018).
- Peng, C., Trojanowski, J. Q. & Lee, V. M. Protein transmission in neurodegenerative disease. *Nat. Rev. Neurol.* **16**, 199–212 (2020).
- Scheckel, C. & Aguzzi, A. Prions, prionoids and protein misfolding disorders. *Nat. Rev. Genet.* **19**, 405–418 (2018).
- Otzen, D. & Riek, R. Functional amyloids. *Cold Spring Harb. Perspect. Biol.* <https://doi.org/10.1101/cshperspect.a033860> (2019).
- Cereghetti, G. et al. Reversible amyloids of pyruvate kinase couple cell metabolism and stress granule disassembly. *Nat. Cell Biol.* **23**, 1085–1094 (2021).
- Jeon, J., Yau, W.-M. & Tycko, R. Early events in amyloid- β self-assembly probed by time-resolved solid state NMR and light scattering. *Nat. Commun.* **14**, 2964 (2023).
- Zimmermann, M. R. et al. Mechanism of secondary nucleation at the single fibril level from direct observations of A β ₄₂ aggregation. *J. Am. Chem. Soc.* **143**, 16621–16629 (2021).
- Wesseling, H. et al. Tau PTM profiles identify patient heterogeneity and stages of Alzheimer's disease. *Cell* **183**, 1699–1713.e13 (2020).
- Tracy, T. E. et al. Tau interactome maps synaptic and mitochondrial processes associated with neurodegeneration. *Cell* <https://doi.org/10.1016/j.cell.2021.12.041> (2022).
- Ferrari, L. et al. Arginine π -stacking drives binding to fibrils of the Alzheimer protein Tau. *Nat. Commun.* **11**, 571 (2020).

17. Gallardo, R., Ranson, N. A. & Radford, S. E. Amyloid structures: much more than just a cross- β fold. *Curr. Opin. Struct. Biol.* **60**, 7–16 (2020).
18. Sawaya, M. R., Hughes, M. P., Rodriguez, J. A., Riek, R. & Eisenberg, D. S. The expanding amyloid family: structure, stability, function, and pathogenesis. *Cell* **184**, 4857–4873 (2021).
19. Louros, N., van der Kant, R., Schymkowitz, J. & Rousseau, F. StAMP-DB: a platform for structures of polymorphic amyloid fibril cores. *Bioinformatics* **38**, 2636–2638 (2022).
20. van der Kant, R., Louros, N., Schymkowitz, J. & Rousseau, F. Thermodynamic analysis of amyloid fibril structures reveals a common framework for stability in amyloid polymorphs. *Structure* <https://doi.org/10.1016/j.str.2022.05.002> (2022).
21. Mullanpudi, V. et al. Network of hotspot interactions cluster tau amyloid folds. *Nat. Commun.* **14**, 895 (2023).
22. Karplus, M. & Weaver, D. L. Protein-folding dynamics. *Nature* **260**, 404–406 (1976).
23. Levinthal, C. Are there pathways for protein folding? *J. Chim. Phys.* **65**, 44–45 (1968).
24. Ellis, R. J. Protein folding: importance of the Anfinsen cage. *Curr. Biol.* **13**, R881–R883 (2003).
25. Anfinsen, C. B. Principles that govern the folding of protein chains. *Science* **181**, 223–230 (1973).
26. Fersht, A. R. & Daggett, V. Protein folding and unfolding at atomic resolution. *Cell* **108**, 573–582 (2002).
27. Kiefhaber, T. Kinetic traps in lysozyme folding. *Proc. Natl Acad. Sci. USA* **92**, 9029–9033 (1995).
28. Silow, M. & Oliveberg, M. Transient aggregates in protein folding are easily mistaken for folding intermediates. *Proc. Natl Acad. Sci. USA* **94**, 6084–6086 (1997).
29. Buchner, J. Molecular chaperones and protein quality control: an introduction to the JBC Reviews thematic series. *J. Biol. Chem.* **294**, 2074–2075 (2019).
30. Powers, E. T., Morimoto, R. I., Dillin, A., Kelly, J. W. & Balch, W. E. Biological and chemical approaches to diseases of proteostasis deficiency. *Annu. Rev. Biochem.* **78**, 959–991 (2009).
31. Baldwin, A. J. et al. Metastability of native proteins and the phenomenon of amyloid formation. *J. Am. Chem. Soc.* **133**, 14160–14163 (2011).
32. Chiti, F. et al. Kinetic partitioning of protein folding and aggregation. *Nat. Struct. Biol.* **9**, 137–143 (2002).
33. Huang, C. et al. Intrinsically aggregation-prone proteins form amyloid-like aggregates and contribute to tissue aging in *Caenorhabditis elegans*. *eLife* <https://doi.org/10.7554/eLife.43059> (2019).
34. Vecchi, G. et al. Proteome-wide observation of the phenomenon of life on the edge of solubility. *Proc. Natl Acad. Sci. USA* **117**, 1015–1020 (2020).
35. Ciryam, P., Kundra, R., Morimoto, R. I., Dobson, C. M. & Vendruscolo, M. Supersaturation is a major driving force for protein aggregation in neurodegenerative diseases. *Trends Pharmacol. Sci.* **36**, 72–77 (2015).
36. Ciryam, P. et al. A metastable subproteome underlies inclusion formation in muscle proteinopathies. *Acta Neuropathol. Commun.* <https://doi.org/10.1186/s40478-019-0853-9> (2019).
37. Niwa, T., Kanamori, T., Ueda, T. & Taguchi, H. Global analysis of chaperone effects using a reconstituted cell-free translation system. *Proc. Natl Acad. Sci. USA* **109**, 8937–8942 (2012).
38. Fujiwara, K., Ishihama, Y., Nakahigashi, K., Soga, T. & Taguchi, H. A systematic survey of in vivo obligate chaperonin-dependent substrates. *EMBO J.* **29**, 1552–1564 (2010).
39. Niwa, T., Fujiwara, K. & Taguchi, H. Identification of novel in vivo obligate GroEL/ES substrates based on data from a cell-free proteomics approach. *FEBS Lett.* **590**, 251–257 (2016).
40. Gestaut, D. et al. Structural visualization of the tubulin folding pathway directed by human chaperonin TRiC/CCT. *Cell* **185**, 4770–4787.e20 (2022).
41. Chen, D. H. et al. Visualizing GroEL/ES in the act of encapsulating a folding protein. *Cell* **153**, 1354–1365 (2013).
42. Sharma, S. et al. Monitoring protein conformation along the pathway of chaperonin-assisted folding. *Cell* **133**, 142–153 (2008).
43. Ellis, R. J. Revisiting the Anfinsen cage. *Fold. Des.* **1**, R9–R15 (1996).
44. Balch, W. E., Morimoto, R. I., Dillin, A. & Kelly, J. W. Adapting proteostasis for disease intervention. *Science* **319**, 916–919 (2008).
45. Sawaya, M. R. et al. Atomic structures of amyloid cross- β spines reveal varied steric zippers. *Nature* **447**, 453–457 (2007).
46. Teng, P. K. & Eisenberg, D. Short protein segments can drive a non-fibrillizing protein into the amyloid state. *Protein Eng. Des. Sel.* **22**, 531–536 (2009).
47. Louros, N. et al. WALTZ-DB 2.0: an updated database containing structural information of experimentally determined amyloid-forming peptides. *Nucleic Acids Res.* **48**, D389–D393 (2020).
48. Santos, J., Pallares, I., Iglesias, V. & Ventura, S. Cryptic amyloidogenic regions in intrinsically disordered proteins: function and disease association. *Comput. Struct. Biotechnol. J.* **19**, 4192–4206 (2021).
49. Fuxreiter, M. Fold or not to fold upon binding—does it really matter? *Curr. Opin. Struct. Biol.* **54**, 19–25 (2019).
50. Oldfield, C. J. & Dunker, A. K. Intrinsically disordered proteins and intrinsically disordered protein regions. *Annu. Rev. Biochem.* **83**, 553–584 (2014).
51. Maurer-Stroh, S. et al. Exploring the sequence determinants of amyloid structure using position-specific scoring matrices. *Nat. Methods* **7**, 237–242 (2010).
52. Louros, N., Orlando, G., De Vleeschouwer, M., Rousseau, F. & Schymkowitz, J. Structure-based machine-guided mapping of amyloid sequence space reveals uncharted sequence clusters with higher solubilities. *Nat. Commun.* **11**, 3314 (2020).
53. Ulapec, S. M., Brockwell, D. J. & Radford, S. E. Looking beyond the core: the role of flanking regions in the aggregation of amyloidogenic peptides and proteins. *Front. Neurosci.* **14**, 611285 (2020).
54. Strodel, B. Energy landscapes of protein aggregation and conformation switching in intrinsically disordered proteins. *J. Mol. Biol.* **433**, 167182 (2021).
55. Mirbaha, H. et al. Inert and seed-competent tau monomers suggest structural origins of aggregation. *eLife* **7**, e36584 (2018).
56. Furukawa, K. et al. Isoelectric point-amyloid formation of α -synuclein extends the generality of the solubility and supersaturation-limited mechanism. *Curr. Res. Struct. Biol.* **2**, 35–44 (2020).
57. Kandola, T. et al. The polyglutamine amyloid nucleus in living cells is a monomer with competing dimensions of order. Preprint at *bioRxiv* <https://doi.org/10.1101/2021.08.29.458132> (2021).
58. Killinger, B. A., Melki, R., Brundin, P. & Kordower, J. H. Endogenous α -synuclein monomers, oligomers and resulting pathology: let's talk about the lipids in the room. *NPJ Parkinsons Dis.* **5**, 23 (2019).
59. Sciacca, M. F. et al. Lipid-chaperone hypothesis: a common molecular mechanism of membrane disruption by intrinsically disordered proteins. *ACS Chem. Neurosci.* **11**, 4336–4350 (2020).
60. Riback, J. A. et al. Innovative scattering analysis shows that hydrophobic disordered proteins are expanded in water. *Science* **358**, 238–241 (2017).
61. Borgia, A. et al. Transient misfolding dominates multidomain protein folding. *Nat. Commun.* **6**, 8861 (2015).
62. Borgia, M. B. et al. Single-molecule fluorescence reveals sequence-specific misfolding in multidomain proteins. *Nature* **474**, 662–665 (2011).
63. Zheng, W., Schafer, N. P. & Wolynes, P. G. Frustration in the energy landscapes of multidomain protein misfolding. *Proc. Natl Acad. Sci. USA* **110**, 1680–1685 (2013).
64. Guthertz, N. et al. The effect of mutation on an aggregation-prone protein: an in vivo, in vitro, and in silico analysis. *Proc. Natl Acad. Sci. USA* **119**, e2200468119 (2022).
65. Ventura, S. et al. Short amino acid stretches can mediate amyloid formation in globular proteins: the Src homology 3 (SH3) case. *Proc. Natl Acad. Sci. USA* **101**, 7258–7263 (2004).
66. Ivanova, M. I., Sawaya, M. R., Gingery, M., Attinger, A. & Eisenberg, D. An amyloid-forming segment of β_2 -microglobulin suggests a molecular model for the fibril. *Proc. Natl Acad. Sci. USA* **101**, 10584–10589 (2004).
67. Ventura, S., Lacroix, E. & Serrano, L. Insights into the origin of the tendency of the PI3-SH3 domain to form amyloid fibrils. *J. Mol. Biol.* **322**, 1147–1158 (2002).
68. Janssen, K. et al. Exploiting the intrinsic misfolding propensity of the KRAS oncoprotein. *Proc. Natl Acad. Sci. USA* **120**, e2214921120 (2023).
69. Michiels, E. et al. Reverse engineering synthetic antiviral amyloids. *Nat. Commun.* **11**, 2832 (2020).
70. Khodaparast, L. et al. Aggregating sequences that occur in many proteins constitute weak spots of bacterial proteostasis. *Nat. Commun.* **9**, 866 (2018).
71. Gallardo, R. et al. De novo design of a biologically active amyloid. *Science* <https://doi.org/10.1126/science.aah4949> (2016).
72. De Baets, G., Van Doorn, L., Rousseau, F. & Schymkowitz, J. Increased aggregation is more frequently associated to human disease-associated mutations than to neutral polymorphisms. *PLoS Comput. Biol.* **11**, e1004374 (2015).
73. Thangakani, A. M., Kumar, S., Velmurugan, D. & Gromiha, M. S. M. How do thermophilic proteins resist aggregation. *Proteins Struct. Funct. Bioinform.* **80**, 1003–1015 (2012).
74. Langenberg, T. et al. Thermodynamic and evolutionary coupling between the native and amyloid state of globular proteins. *Cell Rep.* **31**, 107512 (2020).
75. Geller, R., Pechmann, S., Acevedo, A., Andino, R. & Frydman, J. Hsp90 shapes protein and RNA evolution to balance trade-offs between protein stability and aggregation. *Nat. Commun.* **9**, 1781 (2018).
76. Claes, F. et al. Exposure of a cryptic Hsp70 binding site determines the cytotoxicity of the ALS-associated SOD1-mutant A4V. *Protein Eng. Des. Sel.* **32**, 443–457 (2019).
77. Gilis, D., Massar, S., Cerf, N. J. & Rooman, M. Optimality of the genetic code with respect to protein stability and amino-acid frequencies. *Genome Biol.* **2**, RESEARCH0049 (2001).
78. Buck, P. M., Kumar, S. & Singh, S. K. On the role of aggregation prone regions in protein evolution, stability, and enzymatic catalysis: insights from diverse analyses. *PLoS Comput. Biol.* **9**, e1003291 (2013).
79. Maury, C. P. Self-propagating β -sheet polypeptide structures as prebiotic informational molecular entities: the amyloid world. *Orig. Life Evol. Biosph.* **39**, 141–150 (2009).
80. Carny, O. & Gazit, E. A model for the role of short self-assembled peptides in the very early stages of the origin of life. *FASEB J.* **19**, 1051–1055 (2005).
81. Rout, S. K., Rhyner, D., Riek, R. & Greenwald, J. Prebiotically plausible autocatalytic peptide amyloids. *Chemistry* **28**, e202103841 (2022).
82. Serio, P. R. et al. Nucleated conformational conversion and the replication of conformational information by a prion determinant. *Science* **289**, 1317–1321 (2000).
83. Carny, O. & Gazit, E. Creating prebiotic sanctuary: self-assembling supramolecular peptide structures bind and stabilize RNA. *Orig. Life Evol. Biosph.* **41**, 121–132 (2011).
84. Sanderson, J. M. The association of lipids with amyloid fibrils. *J. Biol. Chem.* **298**, 102108 (2022).
85. Rufo, C. M. et al. Short peptides self-assemble to produce catalytic amyloids. *Nat. Chem.* **6**, 303–309 (2014).
86. Sun-Wang, J. L., Ivanova, S. & Zorzano, A. The dialogue between the ubiquitin-proteasome system and autophagy: implications in ageing. *Ageing Res. Rev.* **64**, 101203 (2020).

87. Balchin, D., Hayer-Hartl, M. & Hartl, F. U. Recent advances in understanding catalysis of protein folding by molecular chaperones. *FEBS Lett.* **594**, 2770–2781 (2020).
88. Mashaghi, A. et al. Alternative modes of client binding enable functional plasticity of Hsp70. *Nature* **539**, 448–451 (2016).
89. Faust, O. et al. HSP40 proteins use class-specific regulation to drive HSP70 functional diversity. *Nature* **587**, 489–494 (2020).
90. Burmann, B. M. et al. Regulation of α -synuclein by chaperones in mammalian cells. *Nature* **577**, 127–132 (2020).
91. Goloubinoff, P., Sassi, A. S., Fauvet, B., Barducci, A. & De Los Rios, P. Chaperones convert the energy from ATP into the nonequilibrium stabilization of native proteins. *Nat. Chem. Biol.* **14**, 388–395 (2018).
92. Kaushik, S. & Cuervo, A. M. The coming of age of chaperone-mediated autophagy. *Nat. Rev. Mol. Cell Biol.* **19**, 365–381 (2018).
93. Xu, C. Y. et al. DJ-1 Inhibits α -synuclein aggregation by regulating chaperone-mediated autophagy. *Front. Aging Neurosci.* **9**, 308 (2017).
94. Bourdenx, M. et al. Chaperone-mediated autophagy prevents collapse of the neuronal metastable proteome. *Cell* **184**, 2696–2714.e25 (2021).
95. Caballero, B. et al. Acetylated tau inhibits chaperone-mediated autophagy and promotes tau pathology propagation in mice. *Nat. Commun.* **12**, 2238 (2021).
96. Baughman, H. E. R., Clouser, A. F., Klevit, R. E. & Nath, A. HspB1 and Hsc70 chaperones engage distinct tau species and have different inhibitory effects on amyloid formation. *J. Biol. Chem.* **293**, 2687–2700 (2018).
97. Kundel, F. et al. Hsp70 inhibits the nucleation and elongation of tau and sequesters tau aggregates with high affinity. *ACS Chem. Biol.* **13**, 636–646 (2018).
98. Avellaneda, M. J. et al. Processive extrusion of polypeptide loops by a Hsp100 disaggregase. *Nature* **578**, 317–320 (2020).
99. Nachman, E. et al. Disassembly of Tau fibrils by the human Hsp70 disaggregation machinery generates small seeding-competent species. *J. Biol. Chem.* **295**, 9676–9690 (2020).
100. Wentink, A. S. et al. Molecular dissection of amyloid disaggregation by human HSP70. *Nature* **587**, 483–488 (2020).
101. Stein, K. C. & Frydman, J. The stop-and-go traffic regulating protein biogenesis: how translation kinetics controls proteostasis. *J. Biol. Chem.* **294**, 2076–2084 (2019).
102. Hanson, G. & Collier, J. Codon optimality, bias and usage in translation and mRNA decay. *Nat. Rev. Mol. Cell Biol.* **19**, 20–30 (2018).
103. Yan, X. W., Hoek, T. A., Vale, R. D. & Tanenbaum, M. E. Dynamics of translation of single mRNA molecules in vivo. *Cell* **165**, 976–989 (2016).
104. Jacobs, W. M. & Shakhnovich, E. I. Evidence of evolutionary selection for cotranslational folding. *Proc. Natl Acad. Sci. USA* **114**, 11434–11439 (2017).
105. Bitran, A., Jacobs, W. M., Zhai, X. & Shakhnovich, E. Cotranslational folding allows misfolding-prone proteins to circumvent deep kinetic traps. *Proc. Natl Acad. Sci. USA* **117**, 1485–1495 (2020).
106. Zhou, T., Weems, M. & Wilke, C. O. Translationally optimal codons associate with structurally sensitive sites in proteins. *Mol. Biol. Evol.* **26**, 1571–1580 (2009).
107. Rosenberg, A. A., Marx, A. & Bronstein, A. M. Codon-specific Ramachandran plots show amino acid backbone conformation depends on identity of the translated codon. *Nat. Commun.* **13**, 2815 (2022).
108. Doring, K. et al. Profiling Ssb-nascent chain interactions reveals principles of Hsp70-assisted folding. *Cell* **170**, 298 (2017).
109. Shen, X., Song, S., Li, C. & Zhang, J. Synonymous mutations in representative yeast genes are mostly strongly non-neutral. *Nature* <https://doi.org/10.1038/s41586-022-04823-w> (2022).
110. Stein, K. C., Morales-Polanco, F., van der Lienden, J., Rainbolt, T. K. & Frydman, J. Ageing exacerbates ribosome pausing to disrupt cotranslational proteostasis. *Nature* **601**, 637–642 (2022).
111. Wruck, F. et al. The ribosome modulates folding inside the ribosomal exit tunnel. *Commun. Biol.* **4**, 523 (2021).
112. Shen, K. et al. Dual role of ribosome-binding domain of NAC as a potent suppressor of protein aggregation and aging-related proteinopathies. *Mol. Cell* **74**, 729–741.e7 (2019).
113. Rousseau, F., Serrano, L. & Schymkowitz, J. W. How evolutionary pressure against protein aggregation shaped chaperone specificity. *J. Mol. Biol.* **355**, 1037–1047 (2006).
114. Monsellier, E. & Chiti, F. Prevention of amyloid-like aggregation as a driving force of protein evolution. *EMBO Rep.* **8**, 737–742 (2007).
115. Houben, B. et al. Autonomous aggregation suppression by acidic residues explains why chaperones favour basic residues. *EMBO J.* <https://doi.org/10.15252/embj.2019102864> (2020).
116. Ganesan, A. et al. Structural hot spots for the solubility of globular proteins. *Nat. Commun.* **7**, 10816 (2016).
117. De Baets, G., Van Durme, J., Rousseau, F. & Schymkowitz, J. A genome-wide sequence-structure analysis suggests aggregation gatekeepers constitute an evolutionary constrained functional class. *J. Mol. Biol.* **426**, 2405–2412 (2014).
118. Casari, G., Sander, C. & Valencia, A. A method to predict functional residues in proteins. *Nat. Struct. Mol. Biol.* **2**, 171–178 (1995).
119. Huang, L. et al. DAXX represents a new type of protein-folding enabler. *Nature* <https://doi.org/10.1038/s41586-021-03824-5> (2021).
120. Doherty, C. P. A. et al. A short motif in the N-terminal region of α -synuclein is critical for both aggregation and function. *Nat. Struct. Mol. Biol.* **27**, 249 (2020).
121. Yoshimura, Y. et al. MOAG-4 promotes the aggregation of α -synuclein by competing with self-protective electrostatic interactions. *J. Biol. Chem.* **292**, 8269–8278 (2017).
122. van Ham, T. J. et al. Identification of MOAG-4/SERF as a regulator of age-related proteotoxicity. *Cell* **142**, 601–612 (2010).
123. Pras, A. et al. The cellular modifier MOAG-4/SERF drives amyloid formation through charge complementation. *EMBO J.* **40**, e107568 (2021).
124. Hong, J.-Y. et al. Structural and dynamic studies reveal that the Ala-rich region of ataxin-7 initiates α -helix formation of the polyQ tract but suppresses its aggregation. *Sci. Rep.* **9**, 7481 (2019).
125. Chen, D. et al. Tau local structure shields an amyloid-forming motif and controls aggregation propensity. *Nat. Commun.* **10**, 2493 (2019).
126. Zhang, W. et al. Modulation of the aggregation of an amyloidogenic sequence by flanking-disordered region in the intrinsically disordered antigen merozoite surface protein 2. *Eur. Biophys. J.* **48**, 99–110 (2019).
127. Santner, A. A. et al. Sweeping away protein aggregation with entropic bristles: intrinsically disordered protein fusions enhance soluble expression. *Biochemistry* **51**, 7250–7262 (2012).
128. Graña-Montes, R., Marinelli, P., Reverter, D. & Ventura, S. N-terminal protein tails act as aggregation protective entropic bristles: the SUMO case. *Biomacromolecules* **15**, 1194–1203 (2014).
129. Michiels, E. et al. Entropic bristles tune the seeding efficiency of prion-nucleating fragments. *Cell Rep.* **30**, 2834–2845.e3 (2020).
130. Housmans, J. A. J. et al. Investigating the sequence determinants of the curling of amyloid fibrils using ovalbumin as a case study. *Biomacromolecules* **23**, 3779–3797 (2022).
131. Shi, Y. et al. Structure-based classification of tauopathies. *Nature* **598**, 359–363 (2021).
132. Iadanza, M. G., Jackson, M. P., Hewitt, E. W., Ranson, N. A. & Radford, S. E. A new era for understanding amyloid structures and disease. *Nat. Rev. Mol. Cell Biol.* **19**, 755–773 (2018).
133. Dueholm, M. S. et al. Fibrillation of the major curli subunit CsgA under a wide range of conditions implies a robust design of aggregation. *Biochemistry* **50**, 8281–8290 (2011).
134. Bemporad, F., De Simone, A., Chiti, F. & Dobson, C. M. Characterizing intermolecular interactions that initiate native-like protein aggregation. *Biophys. J.* **102**, 2595–2604 (2012).
135. Lafita, A., Tian, P., Best, R. B. & Bateman, A. Tandem domain swapping: determinants of multidomain protein misfolding. *Curr. Opin. Struct. Biol.* **58**, 97–104 (2019).
136. Salinas, N. et al. The amphibian antimicrobial peptide uperin 3.5 is a cross- α /cross- β chameleon functional amyloid. *Proc. Natl Acad. Sci. USA* **118**, e2014442118 (2021).
137. Bowden, G. A., Paredes, A. M. & Georgiou, G. Structure and morphology of protein inclusion bodies in *Escherichia coli*. *Biotechnology* **9**, 725–730 (1991).
138. Herneke, A. et al. Protein nanofibrils for sustainable food—characterization and comparison of fibrils from a broad range of plant protein isolates. *ACS Food Sci. Technol.* **1**, 854–864 (2021).
139. Sunde, M. et al. Common core structure of amyloid fibrils by synchrotron X-ray diffraction. *J. Mol. Biol.* **273**, 729–739 (1997).
140. Petkova, A. T. et al. A structural model for Alzheimer's β -amyloid fibrils based on experimental constraints from solid state NMR. *Proc. Natl Acad. Sci. USA* **99**, 16742–16747 (2002).
141. Wasmer, C. et al. Amyloid fibrils of the HET-s(218–289) prion form a β solenoid with a triangular hydrophobic core. *Science* **319**, 1523–1526 (2008).
142. Scheres, S. H., Zhang, W., Falcon, B. & Goedert, M. Cryo-EM structures of tau filaments. *Curr. Opin. Struct. Biol.* **64**, 17–25 (2020).
143. Geddes, A. J., Parker, K. D., Atkins, E. D. & Beighton, E. “Cross- β ” conformation in proteins. *J. Mol. Biol.* **32**, 343–358 (1968).
144. Yang, Y. et al. Cryo-EM structures of amyloid- β_{25} filaments from human brains. *Science* **375**, 167–172 (2022).
145. Fitzpatrick, A. W. P. et al. Cryo-EM structures of tau filaments from Alzheimer's disease. *Nature* **547**, 185–190 (2017).
146. Zhang, W. et al. Novel tau filament fold in corticobasal degeneration. *Nature* **580**, 283–287 (2020).
147. Falcon, B. et al. Tau filaments from multiple cases of sporadic and inherited Alzheimer's disease adopt a common fold. *Acta Neuropathol.* **136**, 699–708 (2018).
148. Schweighauser, M. et al. Structures of α -synuclein filaments from multiple system atrophy. *Nature* **585**, 464–469 (2020).
149. Yang, Y. et al. Structures of α -synuclein filaments from human brains with Lewy pathology. *Nature* **610**, 791–795 (2022).
150. Zhao, K. et al. Parkinson's disease-related phosphorylation at Tyr39 rearranges α -synuclein amyloid fibril structure revealed by cryo-EM. *Proc. Natl Acad. Sci. USA* **117**, 20305–20315 (2020).
151. Guerrero-Ferreira, R., Kovacic, L., Ni, D. & Stahlberg, H. New insights on the structure of α -synuclein fibrils using cryo-electron microscopy. *Curr. Opin. Neurobiol.* **61**, 89–95 (2020).
152. Cao, Q., Boyer, D. R., Sawaya, M. R., Ge, P. & Eisenberg, D. S. Cryo-EM structures of four polymorphic TDP-43 amyloid cores. *Nat. Struct. Mol. Biol.* **26**, 619–627 (2019).
153. Li, Q., Jaroniec, C. P. & Surewicz, W. K. Cryo-EM structure of disease-related prion fibrils provides insights into seeding barriers. *Nat. Struct. Mol. Biol.* **29**, 962–965 (2022).
154. Cao, Q., Boyer, D. R., Sawaya, M. R., Ge, P. & Eisenberg, D. S. Cryo-EM structure and inhibitor design of human IAPP (amylin) fibrils. *Nat. Struct. Mol. Biol.* **27**, 653–659 (2020).
155. Gallardo, R. et al. Fibril structures of diabetes-related amylin variants reveal a basis for surface-templated assembly. *Nat. Struct. Mol. Biol.* **27**, 1048–1056 (2020).

156. Schmidt, M. et al. Cryo-EM structure of a transthyretin-derived amyloid fibril from a patient with hereditary ATTR amyloidosis. *Nat. Commun.* **10**, 5008 (2019).
157. Liberta, F. et al. Cryo-EM fibril structures from systemic AA amyloidosis reveal the species complementarity of pathological amyloids. *Nat. Commun.* **10**, 1104 (2019).
158. Radamaker, L. et al. Cryo-EM structure of a light chain-derived amyloid fibril from a patient with systemic AL amyloidosis. *Nat. Commun.* **10**, 1103 (2019).
159. Kaufman, S. K. et al. Tau prion strains dictate patterns of cell pathology, progression rate, and regional vulnerability in vivo. *Neuron* **92**, 796–812 (2016).
160. Vaquer-Alicea, J., Diamond, M. I. & Joachim, L. A. Tau strains shape disease. *Acta Neuropathol.* **142**, 57–71 (2021).
161. Van der Perren, A. et al. The structural differences between patient-derived α -synuclein strains dictate characteristics of Parkinson's disease, multiple system atrophy and dementia with Lewy bodies. *Acta Neuropathol.* **139**, 977–1000 (2020).
162. Soto, C. & Pritzkow, S. Protein misfolding, aggregation, and conformational strains in neurodegenerative diseases. *Nat. Neurosci.* **21**, 1332–1340 (2018).
163. Maxwell, A. M. et al. Emergence of distinct and heterogeneous strains of amyloid β with advanced Alzheimer's disease pathology in Down syndrome. *Acta Neuropathol. Commun.* **9**, 201 (2021).
164. Abskharon, R. et al. Cryo-EM structure of RNA-induced tau fibrils reveals a small C-terminal core that may nucleate fibril formation. *Proc. Natl Acad. Sci. USA* **119**, e2119952119 (2022).
165. Zhang, W. et al. Heparin-induced tau filaments are polymorphic and differ from those in Alzheimer's and Pick's diseases. *eLife* **8**, e43584 (2019).
166. Tao, Y. et al. Heparin induces α -synuclein to form new fibril polymorphs with attenuated neuropathology. *Nat. Commun.* **13**, 4226 (2022).
167. Herrmann, U. S. et al. Structure-based drug design identifies polythiophenes as antiprion compounds. *Sci. Transl. Med.* **7**, 299ra123 (2015).
168. Heerde, T. et al. Cryo-EM demonstrates the in vitro proliferation of an ex vivo amyloid fibril morphology by seeding. *Nat. Commun.* **13**, 85 (2022).
169. Strohäker, T. et al. Structural heterogeneity of α -synuclein fibrils amplified from patient brain extracts. *Nat. Commun.* **10**, 5535 (2019).
170. Lövestam, S. et al. Seeded assembly in vitro does not replicate the structures of α -synuclein filaments from multiple system atrophy. *FEBS Open Bio* **11**, 999–1013 (2021).
171. Lövestam, S. et al. Assembly of recombinant tau into filaments identical to those of Alzheimer's disease and chronic traumatic encephalopathy. *eLife* **11**, e76494 (2022).
172. Al-Hilaly, Y. K. et al. Tau (297–391) forms filaments that structurally mimic the core of paired helical filaments in Alzheimer's disease brain. *FEBS Lett.* **594**, 944–950 (2020).
173. Radamaker, L. et al. Cryo-EM reveals structural breaks in a patient-derived amyloid fibril from systemic AL amyloidosis. *Nat. Commun.* **12**, 875 (2021).
174. Arakhamia, T. et al. Posttranslational modifications mediate the structural diversity of tauopathy strains. *Cell* **180**, 633–644.e12 (2020).
175. Louros, N. N. et al. Tracking the amyloidogenic core of IAPP amyloid fibrils: insights from micro-Raman spectroscopy. *J. Struct. Biol.* **199**, 140–152 (2017).
176. Ye, X., Hedenqvist, M. S., Langton, M. & Lendel, C. On the role of peptide hydrolysis for fibrillation kinetics and amyloid fibril morphology. *RSC Adv.* **8**, 6915–6924 (2018).
177. Louros, N. et al. Tau amyloid polymorphism is shaped by local structural propensities of its protein sequence. Preprint at *bioRxiv* <https://doi.org/10.1101/2022.10.24.512987> (2022).
178. Michaels, T. C. T. et al. Chemical kinetics for bridging molecular mechanisms and macroscopic measurements of amyloid fibril formation. *Annu. Rev. Phys. Chem.* **69**, 273–298 (2018).
179. Michaels, T. C. T. et al. Fluctuations in the kinetics of linear protein self-assembly. *Phys. Rev. Lett.* **116**, 258103 (2016).
180. Galvagnion, C. et al. Lipid vesicles trigger α -synuclein aggregation by stimulating primary nucleation. *Nat. Chem. Biol.* **11**, 229–234 (2015).
181. Louros, N., Schymkowitz, J. & Rousseau, F. Heterotypic amyloid interactions: clues to polymorphic bias and selective cellular vulnerability? *Curr. Opin. Struct. Biol.* **72**, 176–186 (2022).
182. Juhl, D. W. et al. Conservation of the amyloid interactome across diverse fibrillar structures. *Sci. Rep.* **9**, 3863 (2019).
183. Chatterjee, D. et al. Co-aggregation and secondary nucleation in the life cycle of human prolactin/galanin functional amyloids. *eLife* **11**, e73835 (2022).
184. Pansieri, J. et al. Templating S100A9 amyloids on $\alpha\beta$ fibrillar surfaces revealed by charge detection mass spectrometry, microscopy, kinetic and microfluidic analyses. *Chem. Sci.* **11**, 7031–7039 (2020).
185. Linse, S. Monomer-dependent secondary nucleation in amyloid formation. *Biophys. Rev.* **9**, 329–338 (2017).
186. Törnqvist, M. & Linse, S. Chiral selectivity of secondary nucleation in amyloid fibril propagation. *Angew. Chem. Int. Ed. Engl.* **60**, 24008–24011 (2021).
187. Thacker, D. et al. The role of fibril structure and surface hydrophobicity in secondary nucleation of amyloid fibrils. *Proc. Natl Acad. Sci. USA* **117**, 25272–25283 (2020).
188. Louros, N. et al. Mapping the sequence specificity of heterotypic amyloid interactions enables the identification of aggregation modifiers. *Nat. Commun.* **13**, 1351 (2022).
189. Tian, Y. & Viles, J. H. pH dependence of amyloid- β fibril assembly kinetics: unravelling the microscopic molecular processes. *Angew. Chem. Int. Ed.* **61**, e202210675 (2022).
190. Kumari, P. et al. Structural insights into α -synuclein monomer–fibril interactions. *Proc. Natl Acad. Sci. USA* **118**, e2012171118 (2021).
191. Hadi Alijanvand, S., Peduzzo, A. & Buell, A. K. Secondary nucleation and the conservation of structural characteristics of amyloid fibril strains. *Front. Mol. Biosci.* <https://doi.org/10.3389/fmolb.2021.669994> (2021).
192. Brännström, K. et al. The properties of amyloid- β fibrils are determined by their path of formation. *J. Mol. Biol.* **430**, 1940–1949 (2018).
193. Sakalauskas, A., Ziaunys, M. & Smirnovas, V. Concentration-dependent polymorphism of insulin amyloid fibrils. *PeerJ* **7**, e8208 (2019).
194. Wördehoff, M. M. et al. Single fibril growth kinetics of α -synuclein. *J. Mol. Biol.* **427**, 1428–1435 (2015).
195. Sleutel, M. et al. Nucleation and growth of a bacterial functional amyloid at single-fiber resolution. *Nat. Chem. Biol.* **13**, 902–908 (2017).
196. Sun, Y. et al. Direct observation of competing prion protein fibril populations with distinct structures and kinetics. *ACS Nano* <https://doi.org/10.1021/acsnano.2c12009> (2023).
197. Pinotsi, D. et al. Direct observation of heterogeneous amyloid fibril growth kinetics via two-color super-resolution microscopy. *Nano Lett.* **14**, 339–345 (2014).
198. Rodriguez, R. A., Chen, L. Y., Plascencia-Villa, G. & Perry, G. Thermodynamics of amyloid- β fibril elongation: atomistic details of the transition state. *ACS Chem. Neurosci.* **9**, 783–789 (2018).
199. Sasmal, S., Schwierz, N. & Head-Gordon, T. Mechanism of nucleation and growth of $A\beta_{40}$ fibrils from all-atom and coarse-grained simulations. *J. Phys. Chem. B* **120**, 12088–12097 (2016).
200. Xu, Y. et al. Frustrated peptide chains at the fibril tip control the kinetics of growth of amyloid- β fibrils. *Proc. Natl Acad. Sci. USA* **118**, e2110995118 (2021).
201. Xu, Y. et al. Steady, symmetric, and reversible growth and dissolution of individual amyloid- β fibrils. *ACS Chem. Neurosci.* **10**, 2967–2976 (2019).
202. Nicoud, L., Lazzari, S., Balderas Barragán, D. & Morbidelli, M. Fragmentation of amyloid fibrils occurs in preferential positions depending on the environmental conditions. *J. Phys. Chem. B* **119**, 4644–4652 (2015).
203. Sanami, S., Purton, T. J., Smith, D. P., Tuite, M. F. & Xue, W. F. Comparative analysis of the relative fragmentation stabilities of polymorphic α -synuclein amyloid fibrils. *Biomolecules* <https://doi.org/10.3390/biom12050630> (2022).
204. Beal, D. M. et al. The division of amyloid fibrils: systematic comparison of fibril fragmentation stability by linking theory with experiments. *iScience* **23**, 101512 (2020).
205. Xue, W.-F., Hellewell, A. L., Hewitt, E. W. & Radford, S. E. Fibril fragmentation in amyloid assembly and cytotoxicity. *Prion* **4**, 20–25 (2010).
206. Zhang, X. et al. Correlation between cellular uptake and cytotoxicity of fragmented α -synuclein amyloid fibrils suggests intracellular basis for toxicity. *ACS Chem. Neurosci.* **11**, 233–241 (2020).
207. Morris, R. J. et al. Mechanistic and environmental control of the prevalence and lifetime of amyloid oligomers. *Nat. Commun.* **4**, 1891 (2013).
208. Konstantouleas, K. et al. Heterotypic amyloid β interactions facilitate amyloid assembly and modify amyloid structure. *EMBO J.* **41**, e108591 (2022).
209. Dear, A. J. et al. Kinetic diversity of amyloid oligomers. *Proc. Natl Acad. Sci. USA* **117**, 12087–12094 (2020).
210. Michaels, T. C. T. et al. Dynamics of oligomer populations formed during the aggregation of Alzheimer's $A\beta_{42}$ peptide. *Nat. Chem.* **12**, 445–451 (2020).
211. Crespo, R. et al. What can the kinetics of amyloid fibril formation tell about off-pathway aggregation? *J. Biol. Chem.* **291**, 2018–2032 (2016).
212. Hasecke, F. et al. Origin of metastable oligomers and their effects on amyloid fibril self-assembly. *Chem. Sci.* **9**, 5937–5948 (2018).
213. Hasecke, F. et al. Protofibril–fibril interactions inhibit amyloid fibril assembly by obstructing secondary nucleation. *Angew. Chem. Int. Ed.* **60**, 3016–3021 (2021).
214. Konno, H. et al. Dynamics of oligomer and amyloid fibril formation by yeast prion Sup35 observed by high-speed atomic force microscopy. *Proc. Natl Acad. Sci. USA* **117**, 7831–7836 (2020).
215. Kaye, R. & Lasagna-Reeves, C. A. Molecular mechanisms of amyloid oligomers toxicity. *J. Alzheimers Dis.* **33**, S67–S78 (2013).
216. He, Y. et al. Amyloid β oligomers suppress excitatory transmitter release via presynaptic depletion of phosphatidylinositol-4,5-bisphosphate. *Nat. Commun.* **10**, 1193 (2019).
217. Lasagna-Reeves, C. A. et al. Tau oligomers impair memory and induce synaptic and mitochondrial dysfunction in wild-type mice. *Mol. Neurodegener.* **6**, 39 (2011).
218. Cascella, R., Bigi, A., Cremades, N. & Cecchi, C. Effects of oligomer toxicity, fibril toxicity and fibril spreading in synucleinopathies. *Cell. Mol. Life Sci.* **79**, 174 (2022).
219. Wagner, J. et al. Medin co-aggregates with vascular amyloid- β in Alzheimer's disease. *Nature* **612**, 123–131 (2022).
220. Konstantouleas, K., Louros, N., Rousseau, F. & Schymkowitz, J. Heterotypic interactions in amyloid function and disease. *FEBS J.* **289**, 2025–2046 (2022).
221. Clinton, L. K., Blurton-Jones, M., Myczek, K., Trojanowski, J. Q. & LaFerla, F. M. Synergistic interactions between $A\beta$, tau, and α -synuclein: acceleration of neuropathology and cognitive decline. *J. Neurosci.* **30**, 7281–7289 (2010).
222. Zhao, Y. et al. β_2 -Microglobulin coaggregates with $A\beta$ and contributes to amyloid pathology and cognitive deficits in Alzheimer's disease model mice. *Nat. Neurosci.* <https://doi.org/10.1038/s41593-023-01352-1> (2023).
223. Yu, A. et al. Tau protein aggregates inhibit the protein-folding and vesicular trafficking arms of the cellular proteostasis network. *J. Biol. Chem.* **294**, 7917–7930 (2019).
224. Engstrom, A. K. et al. The inhibition of LSD1 via sequestration contributes to tau-mediated neurodegeneration. *Proc. Natl Acad. Sci. USA* **117**, 29133–29143 (2020).
225. Peng, C. et al. Cellular milieu imparts distinct pathological α -synuclein strains in α -synucleinopathies. *Nature* **557**, 558–563 (2018).
226. Lengyel, Z., Rufo, C. M. & Korendovych, I. V. Preparation and screening of catalytic amyloid assemblies. *Methods Mol. Biol.* **1777**, 261–270 (2018).

227. Dolan, M. A. et al. Catalytic nanoassemblies formed by short peptides promote highly enantioselective transfer hydrogenation. *ACS Nano* **13**, 9292–9297 (2019).
228. Arad, E., Baruch Leshem, A., Rapaport, H. & Jelinek, R. β -Amyloid fibrils catalyze neurotransmitter degradation. *Chem. Catal.* **1**, 908–922 (2021).
229. Kollmer, M. et al. Electron tomography reveals the fibril structure and lipid interactions in amyloid deposits. *Proc. Natl Acad. Sci. USA* **113**, 5604–5609 (2016).
230. Bäuerlein, F. J. B. et al. In situ architecture and cellular interactions of PolyQ inclusions. *Cell* **171**, 179–187.e10 (2017).
231. Seidler, P. M. et al. Structure-based inhibitors of tau aggregation. *Nat. Chem.* **10**, 170–176 (2018).
232. Agerschow, E. D. et al. An engineered monomer binding-protein for α -synuclein efficiently inhibits the proliferation of amyloid fibrils. *eLife* <https://doi.org/10.7554/eLife.46112> (2019).
233. Lu, J. et al. Structure-based peptide inhibitor design of amyloid- β aggregation. *Front. Mol. Neurosci.* **12**, 54 (2019).
234. Saellies, L. et al. A pair of peptides inhibits seeding of the hormone transporter transthyretin into amyloid fibrils. *J. Biol. Chem.* **294**, 6130–6141 (2019).
235. Krotee, P. et al. Common fibrillar spines of amyloid- β and human islet amyloid polypeptide revealed by microelectron diffraction and structure-based inhibitors. *J. Biol. Chem.* **293**, 2888–2902 (2018).
236. Mompéan, M. et al. The structure of the necrosome RIPK1–RIPK3 core, a human hetero-amyloid signaling complex. *Cell* **173**, 1244–1253.e10 (2018).
237. Cerofolini, L. et al. Mixing A β_{1-40} and A β_{1-42} peptides generates unique amyloid fibrils. *Chem. Commun.* **56**, 8830–8833 (2020).
238. Ziaunys, M., Mikalauskaitė, K. & Smirnovas, V. Amyloidophilic molecule interactions on the surface of insulin fibrils: cooperative binding and fluorescence quenching. *Sci. Rep.* **9**, 20303 (2019).
239. Fichou, Y. et al. Tau-cofactor complexes as building blocks of tau fibrils. *Front. Neurosci.* **13**, 1339 (2019).
240. Yamaguchi, K. et al. Polyphosphates induce amyloid fibril formation of α -synuclein in concentration-dependent distinct manners. *J. Biol. Chem.* **296**, 100510 (2021).
241. Matafora, V. et al. Amyloid aggregates accumulate in melanoma metastasis modulating YAP activity. *EMBO Rep.* **21**, e50446 (2020).
242. Wegmann, S., Medalsky, I. D., Mandelkow, E. & Müller, D. J. The fuzzy coat of pathological human Tau fibrils is a two-layered polyelectrolyte brush. *Proc. Natl Acad. Sci. USA* **110**, E313–E321 (2013).
243. Khan, M. R. et al. Amyloidogenic oligomerization transforms *Drosophila* Orb2 from a translation repressor to an activator. *Cell* **163**, 1468–1483 (2015).
244. Wentink, A., Nussbaum-Krammer, C. & Bukau, B. Modulation of amyloid states by molecular chaperones. *Cold Spring Harb. Perspect. Biol.* <https://doi.org/10.1101/cshperspect.a033969> (2019).
245. Ayyadevara, S., Ganne, A., Balasubramaniam, M. & Shmookler Reis, R. J. Intrinsically disordered proteins identified in the aggregate proteome serve as biomarkers of neurodegeneration. *Metab. Brain Dis.* **37**, 147–152 (2022).
246. Jiang, L. L., Guan, W. L., Wang, J. Y., Zhang, S. X. & Hu, H. Y. RNA-assisted sequestration of RNA-binding proteins by cytoplasmic inclusions of the C-terminal 35-kDa fragment of TDP-43. *J. Cell Sci.* **135**, jcs259380 (2022).
247. Niss, F., Zaidi, W., Hallberg, E. & Ström, A. L. Polyglutamine expanded Ataxin-7 induces DNA damage and alters FUS localization and function. *Mol. Cell Neurosci.* **110**, 103584 (2021).
248. Nihei, Y. et al. Poly-glycine-alanine exacerbates C9orf72 repeat expansion-mediated DNA damage via sequestration of phosphorylated ATM and loss of nuclear hnRNP A3. *Acta Neuropathol.* **139**, 99–118 (2020).
249. Agarwal, A., Arora, L., Rai, S. K., Avni, A. & Mukhopadhyay, S. Spatiotemporal modulations in heterotypic condensates of prion and α -synuclein control phase transitions and amyloid conversion. *Nat. Commun.* **13**, 1154 (2022).
250. Mathieu, C., Pappu, R. V. & Taylor, J. P. Beyond aggregation: pathological phase transitions in neurodegenerative disease. *Science* **370**, 56–60 (2020).
251. Yue, H.-W., Hong, J.-Y., Zhang, S.-X., Jiang, L.-L. & Hu, H.-Y. PolyQ-expanded proteins impair cellular proteostasis of ataxin-3 through sequestering the co-chaperone HSP1 into aggregates. *Sci. Rep.* **11**, 7815 (2021).
252. Ferdosh, S., Banerjee, S., Singh, J. & Barat, C. Amyloid protein-induced sequestration of the eukaryotic ribosome: effect of stoichiometry and polyphenolic inhibitors. *FEBS Lett.* **596**, 1190–1202 (2022).
253. Guo, Q. et al. In situ structure of neuronal C9orf72 poly-GA aggregates reveals proteasome recruitment. *Cell* **172**, 696–705.e12 (2018).
254. Riemenschneider, H. et al. Gel-like inclusions of C-terminal fragments of TDP-43 sequester stalled proteasomes in neurons. *EMBO Rep.* **23**, e53890 (2022).
255. Thibaut, T. A., Anderson, R. T. & Smith, D. M. A common mechanism of proteasome impairment by neurodegenerative disease-associated oligomers. *Nat. Commun.* **9**, 1097 (2018).
256. Riera-Tur, I. et al. Amyloid-like aggregating proteins cause lysosomal defects in neurons via gain-of-function toxicity. *Life Sci. Alliance* **5**, e202101185 (2022).
257. Koss, D. J. et al. RAB39B is redistributed in dementia with Lewy bodies and is sequestered within $\alpha\beta$ plaques and Lewy bodies. *Brain Pathol.* **31**, 120–132 (2021).
258. Morten, M. J. et al. Quantitative super-resolution imaging of pathological aggregates reveals distinct toxicity profiles in different synucleinopathies. *Proc. Natl Acad. Sci. USA* **119**, e2205591119 (2022).
259. Suzuki, G. et al. α -Synuclein strains that cause distinct pathologies differentially inhibit proteasome. *eLife* **9**, e56825 (2020).
260. Mahul-Mellier, A.-L. et al. The process of Lewy body formation, rather than simply α -synuclein fibrillization, is one of the major drivers of neurodegeneration. *Proc. Natl Acad. Sci. USA* **117**, 4971–4982 (2020).
261. Man, W. K. et al. The docking of synaptic vesicles on the presynaptic membrane induced by α -synuclein is modulated by lipid composition. *Nat. Commun.* **12**, 927 (2021).
262. Choong, C.-J. et al. Phosphatidylinositol-3,4,5-trisphosphate interacts with α -synuclein and initiates its aggregation and formation of Parkinson's disease-related fibril polymorphism. *Acta Neuropathol.* <https://doi.org/10.1007/s00401-023-02555-3> (2023).
263. Das, M. & Gursky, O. Amyloid-forming properties of human apolipoproteins: sequence analyses and structural insights. *Adv. Exp. Med. Biol.* **855**, 175–211 (2015).
264. Louros, N. N. et al. Chameleon 'aggregation-prone' segments of apoA-I: a model of amyloid fibrils formed in apoA-I amyloidosis. *Int. J. Biol. Macromol.* **79**, 711–718 (2015).
265. Kenyaga, J. M., Cheng, Q. & Qiang, W. Early stage β -amyloid-membrane interactions modulate lipid dynamics and influence structural interfaces and fibrillation. *J. Biol. Chem.* **298**, 102491 (2022).
266. Antonschmidt, L. et al. Insights into the molecular mechanism of amyloid filament formation: segmental folding of α -synuclein on lipid membranes. *Sci. Adv.* **7**, eabg2174 (2021).
267. Korshavn, K. J. et al. Reduced lipid bilayer thickness regulates the aggregation and cytotoxicity of amyloid- β . *J. Biol. Chem.* **292**, 4638–4650 (2017).
268. Sienski, G. et al. APOE4 disrupts intracellular lipid homeostasis in human iPSC-derived glia. *Sci. Transl. Med.* <https://doi.org/10.1126/scitranslmed.aaz4564> (2021).
269. Blanchard, J. W. et al. APOE4 impairs myelination via cholesterol dysregulation in oligodendrocytes. *Nature* **611**, 769–779 (2022).
270. Liu, C. C. et al. ApoE4 accelerates early seeding of amyloid pathology. *Neuron* **96**, 1024–1032.e23 (2017).
271. Verghese, P. B. et al. ApoE influences amyloid- β (A β) clearance despite minimal apoE/A β association in physiological conditions. *Proc. Natl Acad. Sci. USA* **110**, E1807–E1816 (2013).
272. Wu, J. et al. Cryo-electron microscopy imaging of Alzheimer's amyloid- β_{42} oligomer displayed on a functionally and structurally relevant scaffold. *Angew. Chem. Int. Ed.* **60**, 18680–18687 (2021).
273. Laganowsky, A. et al. Atomic view of a toxic amyloid small oligomer. *Science* **335**, 1228–1231 (2012).
274. Chen, S. W. et al. Structural characterization of toxic oligomers that are kinetically trapped during α -synuclein fibril formation. *Proc. Natl Acad. Sci. USA* **112**, E1994–E2003 (2015).
275. Fusco, G. et al. Structural basis of membrane disruption and cellular toxicity by α -synuclein oligomers. *Science* **358**, 1440–1443 (2017).
276. De, S. et al. Different soluble aggregates of A β_{25} can give rise to cellular toxicity through different mechanisms. *Nat. Commun.* **10**, 1541 (2019).
277. Birol, M., Kumar, S., Rhoades, E. & Miranker, A. D. Conformational switching within dynamic oligomers underpins toxic gain-of-function by diabetes-associated amyloid. *Nat. Commun.* **9**, 1312 (2018).
278. Elenbaas, B. O. W., Kremsreiter, S. M., Khemtremourian, L., Killian, J. A. & Sinnige, T. Fibril elongation by human islet amyloid polypeptide is the main event linking aggregation to membrane damage. *BBA Adv.* **3**, 100083 (2023).
279. Koike, H. & Katsuno, M. The ultrastructure of tissue damage by amyloid fibrils. *Molecules* <https://doi.org/10.3390/molecules26154611> (2021).
280. Friege, B. et al. The 3D structure of lipidic fibrils of α -synuclein. *Nat. Commun.* **13**, 6810 (2022).
281. Shahmoradian, S. H. et al. Lewy pathology in Parkinson's disease consists of crowded organelles and lipid membranes. *Nat. Neurosci.* **22**, 1099–1109 (2019).
282. Lee, J.-E. et al. Mapping surface hydrophobicity of α -synuclein oligomers at the nanoscale. *Nano Lett.* **18**, 7494–7501 (2018).
283. Hachchi, J. et al. Cholesterol catalyzes A β_{42} aggregation through a heterogeneous nucleation pathway in the presence of lipid membranes. *Nat. Chem.* **10**, 673–683 (2018).
284. Limbocker, R. et al. Trodusquemine enhances A β_{42} aggregation but suppresses its toxicity by displacing oligomers from cell membranes. *Nat. Commun.* **10**, 225 (2019).
285. Ganz, T. et al. Microbial pathogens induce neurodegeneration in Alzheimer's disease mice: protection by microglial regulation. *J. Neuroinflammation* **19**, 5 (2022).
286. Brown, G. C. The endotoxin hypothesis of neurodegeneration. *J. Neuroinflammation* **16**, 180 (2019).
287. Yao, L. et al. Protective effects of endotoxin tolerance on peripheral lipopolysaccharide-induced neuroinflammation and dopaminergic neuronal injury. *Immunopharmacol. Immunotoxicol.* **44**, 326–337 (2022).
288. Kim, C. et al. Exposure to bacterial endotoxin generates a distinct strain of α -synuclein fibril. *Sci. Rep.* **6**, 30891 (2016).
289. Cheng, N., Liang, Y., Du, X. & Ye, R. D. Serum amyloid A promotes LPS clearance and suppresses LPS-induced inflammation and tissue injury. *EMBO Rep.* **19**, e45517 (2018).
290. Cammann, D. et al. Genetic correlations between Alzheimer's disease and gut microbiome genera. *Sci. Rep.* **13**, 5258 (2023).
291. Wallen, Z. D. et al. Metagenomics of Parkinson's disease implicates the gut microbiome in multiple disease mechanisms. *Nat. Commun.* **13**, 6958 (2022).
292. Wang, X. et al. Sodium oligomannate therapeutically remodels gut microbiota and suppresses gut bacterial amino acids-shaped neuroinflammation to inhibit Alzheimer's disease progression. *Cell Res.* **29**, 787–803 (2019).

293. Claesson, M. J. et al. Gut microbiota composition correlates with diet and health in the elderly. *Nature* **488**, 178–184 (2012).
294. Seo, D. O. et al. ApoE isoform- and microbiota-dependent progression of neurodegeneration in a mouse model of tauopathy. *Science* **379**, eadd1236 (2023).
295. Sampson, T. R. et al. A gut bacterial amyloid promotes α -synuclein aggregation and motor impairment in mice. *eLife* **9**, e53111 (2020).
296. Perov, S. et al. Structural insights into curli CsgA cross- β fibril architecture inspire repurposing of anti-amyloid compounds as anti-biofilm agents. *PLoS Pathog.* **15**, e1007978 (2019).
297. Bhoite, S. S., Han, Y., Ruotolo, B. T. & Chapman, M. R. Mechanistic insights into accelerated α -synuclein aggregation mediated by human microbiome-associated functional amyloids. *J. Biol. Chem.* **298**, 102088 (2022).
298. Evans, M. L. et al. The bacterial curli system possesses a potent and selective inhibitor of amyloid formation. *Mol. Cell* **57**, 445–455 (2015).
299. Kumar, D. K. et al. Amyloid- β peptide protects against microbial infection in mouse and worm models of Alzheimer's disease. *Sci. Transl. Med.* **8**, 340ra372 (2016).
300. Bückner, R. et al. The cryo-EM structures of two amphibian antimicrobial cross- β amyloid fibrils. *Nat. Commun.* **13**, 4356 (2022).
301. Pastore, A., Raimondi, F., Rajendran, L. & Temussi, P. A. Why does the A β peptide of Alzheimer share structural similarity with antimicrobial peptides? *Commun. Biol.* **3**, 135 (2020).
302. Tang, Y., Zhang, D., Gong, X. & Zheng, J. Repurposing of intestinal defensins as multi-target, dual-function amyloid inhibitors via cross-seeding. *Chem. Sci.* **13**, 7143–7156 (2022).
303. Colombo, A. V. et al. Microbiota-derived short chain fatty acids modulate microglia and promote A β plaque deposition. *eLife* **10**, e59826 (2021).
304. Sade, D., Shaham-Niv, S., Arnon, Z. A., Tavassoly, O. & Gazit, E. Seeding of proteins into amyloid structures by metabolite assemblies may clarify certain unexplained epidemiological associations. *Open Biol.* <https://doi.org/10.1098/rsob.170229> (2018).
305. Tavassoly, O. et al. Quinolinic acid amyloid-like fibrillar assemblies seed α -synuclein aggregation. *J. Mol. Biol.* **430**, 3847–3862 (2018).
306. Zhu, G., Zhao, J., Zhang, H., Wang, G. & Chen, W. Gut microbiota and its metabolites: bridge of dietary nutrients and Alzheimer's disease. *Adv. Nutr.* **14**, 819–839 (2023).
307. Rising, A. et al. AA amyloid in human food chain is a possible biohazard. *Sci. Rep.* **11**, 21069 (2021).
308. Rahman, M. M. et al. Food protein-derived amyloids do not accelerate amyloid β aggregation. *Sci. Rep.* **13**, 985 (2023).
309. Vaneyck, J., Segers-Nolten, I., Broersen, K. & Claessens, M. M. A. E. Cross-seeding of α -synuclein aggregation by amyloid fibrils of food proteins. *J. Biol. Chem.* **296**, 100358 (2021).
310. Monge-Morera, M. et al. Heating wheat gluten promotes the formation of amyloid-like fibrils. *ACS Omega* **6**, 1823–1833 (2021).
311. Lambrecht, M. A. et al. Hydrothermal treatments cause wheat gluten-derived peptides to form amyloid-like fibrils. *J. Agric. Food Chem.* **69**, 1963–1974 (2021).
312. Lambrecht, M. A. et al. Conditions governing food protein amyloid fibril formation. Part II: milk and legume proteins. *Compr. Rev. Food Sci. Food Saf.* **18**, 1277–1291 (2019).
313. Jansens, K. J. A. et al. Conditions governing food protein amyloid fibril formation—part I: egg and cereal proteins. *Compr. Rev. Food Sci. Food Saf.* **18**, 1256–1276 (2019).
314. Cao, Y. & Mezzenga, R. Food protein amyloid fibrils: origin, structure, formation, characterization, applications and health implications. *Adv. Colloid Interface Sci.* **269**, 334–356 (2019).
315. Ye, X. et al. High-temperature and chemically resistant foams from sustainable nanostructured protein. *Adv. Sustain. Syst.* <https://doi.org/10.1002/advs.202100063> (2021).
316. Ezzat, K. et al. The viral protein corona directs viral pathogenesis and amyloid aggregation. *Nat. Commun.* **10**, 2331 (2019).
317. Yu, H. & Wu, J. Amyloid- β : a double agent in Alzheimer's disease? *Biomed. Pharmacother.* **139**, 111575 (2021).
318. Semerdzhiev, S. A., Fakhree, M. A. A., Segers-Nolten, I., Blum, C. & Claessens, M. M. A. E. Interactions between SARS-CoV-2 N-protein and α -synuclein accelerate amyloid formation. *ACS Chem. Neurosci.* **13**, 143–150 (2022).
319. Bhardwaj, T. et al. Amyloidogenic proteins in the SARS-CoV and SARS-CoV-2 proteomes. *Nat. Commun.* **14**, 945 (2023).
320. Nyström, S. & Hammarström, P. Amyloidogenesis of SARS-CoV-2 spike protein. *J. Am. Chem. Soc.* **144**, 8945–8950 (2022).
321. Nilsson, J. F. et al. Molecular determinants of fibrillation in a viral amyloidogenic domain from combined biochemical and biophysical studies. *Int. J. Mol. Sci.* **24** (2023).
322. Leblanc, P. & Vorberg, I. M. Viruses in neurodegenerative diseases: more than just suspects in crimes. *PLoS Pathog.* **18**, e1010670 (2022).
323. Liu, S. et al. Highly efficient intercellular spreading of protein misfolding mediated by viral ligand–receptor interactions. *Nat. Commun.* **12**, 5739 (2021).
324. Damy, T. et al. Efficacy and safety of tafamidis doses in the Tafamidis in Transthyretin Cardiomyopathy Clinical Trial (ATTR-ACT) and long-term extension study. *Eur. J. Heart Fail.* **23**, 277–285 (2021).
325. Torres, L. & Conran, N. Emerging pharmacotherapeutic approaches for the management of sickle cell disease. *Expert. Opin. Pharmacother.* **20**, 173–186 (2019).
326. van Dyck, C. H. et al. Lecanemab in early Alzheimer's disease. *N. Engl. J. Med.* **388**, 9–21 (2022).
327. Mullard, A. Anti-tau antibody failures stack up. *Nat. Rev. Drug Discov.* **20**, 888 (2021).
328. Pillar, C. Blots on a field? *Science* **377**, 358–363 (2022).
329. Niu, Z. et al. Structural insight into IAPP-derived amyloid inhibitors and their mechanism of action. *Angew. Chem. Int. Ed. Engl.* **59**, 5771–5781 (2020).
330. Perchiacca, J. M., Ladiwala, A. R., Bhattacharya, M. & Tessier, P. M. Structure-based design of conformation- and sequence-specific antibodies against amyloid β . *Proc. Natl Acad. Sci. USA* **109**, 84–89 (2012).
331. Limbicker, R. et al. Rationally designed antibodies as research tools to study the structure-toxicity relationship of amyloid- β oligomers. *Int. J. Mol. Sci.* <https://doi.org/10.3390/ijms21124542> (2020).
332. Aprile, F. A. et al. Rational design of a conformation-specific antibody for the quantification of A β oligomers. *Proc. Natl Acad. Sci. USA* **117**, 13509–13518 (2020).
333. Taglialegna, A., Lasa, I. & Valle, J. Amyloid structures as biofilm matrix scaffolds. *J. Bacteriol.* **198**, 2579–2588 (2016).
334. Levkovich, S. A., Gazit, E. & Laor Bar-Yosef, D. Two decades of studying functional amyloids in microorganisms. *Trends Microbiol.* **29**, 251–265 (2021).
335. Louros, N. N. et al. A common 'aggregation-prone' interface possibly participates in the self-assembly of human zona pellucida proteins. *FEBS Lett.* **590**, 619–630 (2016).
336. Louros, N. N., Iconomidou, V. A., Giannelou, P. & Hamodrakas, S. J. Structural analysis of peptide-analogues of human zona pellucida ZP1 protein with amyloidogenic properties: insights into mammalian zona pellucida formation. *PLoS ONE* **8**, e73258 (2013).
337. Louros, N. N. et al. Structural studies of "aggregation-prone" peptide-analogues of teleostean egg chorion ZPB proteins. *Biopolymers* **102**, 427–436 (2014).
338. Maji, S. K. et al. Functional amyloids as natural storage of peptide hormones in pituitary secretory granules. *Science* **325**, 328–332 (2009).
339. Nasi, G. I. et al. *Arabidopsis thaliana* plant natriuretic peptide active domain forms amyloid-like fibrils in a pH-dependent manner. *Plants* <https://doi.org/10.3390/plants1010009> (2021).
340. Wickner, R. B. et al. Yeast prions: structure, biology, and prion-handling systems. *Microbiol. Mol. Biol. Rev.* **79**, 1–17 (2015).
341. Si, K., Lindquist, S. & Kandel, E. R. A neuronal isoform of the aplysia CPEB has prion-like properties. *Cell* **115**, 879–891 (2003).
342. Kajava, A. V. & Steven, A. C. β -Rolls, β -helices, and other β -solenoid proteins. *Adv. Protein Chem.* **73**, 55–96 (2006).
343. Louros, N. N., Baltoumas, F. A., Hamodrakas, S. J. & Iconomidou, V. A. A β -solenoid model of the Pmel17 repeat domain: insights to the formation of functional amyloid fibrils. *J. Comput. Aided Mol. Des.* **30**, 153–164 (2016).
344. Van Melckebeke, H. et al. Atomic-resolution three-dimensional structure of HET-s(218–289) amyloid fibrils by solid-state NMR spectroscopy. *J. Am. Chem. Soc.* **132**, 13765–13775 (2010).
345. Tsiolaki, P. L., Louros, N. N. & Iconomidou, V. A. Hexapeptide tandem repeats dictate the formation of silkworm chorion, a natural protective amyloid. *J. Mol. Biol.* **430**, 3774–3783 (2018).
346. Louros, N. N. & Iconomidou, V. A. Identification of an amyloid fibril forming segment of human Pmel17 repeat domain (RPT domain). *Biopolymers* **106**, 133–139 (2016).
347. Sleutel, M., Pradhan, B. & Remaut, H. Structural analysis of the bacterial amyloid curli. *Nat. Commun.* **14**, 2822 (2022).
348. Louros, N. N., Bolas, G. M. P., Tsiolaki, P. L., Hamodrakas, S. J. & Iconomidou, V. A. Intrinsic aggregation propensity of the CsgB nucleator protein is crucial for curli fiber formation. *J. Struct. Biol.* **195**, 179–189 (2016).
349. Ragonis-Bachar, P. & Landau, M. Functional and pathological amyloid structures in the eyes of 2020 cryo-EM. *Curr. Opin. Struct. Biol.* **68**, 184–193 (2021).
350. Garcia-Pardo, J. et al. Cryo-EM structure of hnRNPL2 fibrils, a functional amyloid associated with limb-girdle muscular dystrophy D3. *Nat. Commun.* **14**, 239 (2023).
351. Lu, J. et al. Cryo-EM structure of the low-complexity domain of hnRNPA2 and its conversion to pathogenic amyloid. *Nat. Commun.* **11**, 4090 (2020).
352. Hervas, R. et al. Cryo-EM structure of a neuronal functional amyloid implicated in memory persistence in *Drosophila*. *Science* **367**, 1230–1234 (2020).
353. Seuring, C. et al. The three-dimensional structure of human β -endorphin amyloid fibrils. *Nat. Struct. Mol. Biol.* **27**, 1178–1184 (2020).
354. Liu, Y., Liu, J. & He, X. Different pK_a shifts of internal GLU8 in human β -endorphin amyloid revealing a coupling of internal ionization and stepwise fibril disassembly. *J. Phys. Chem. B* <https://doi.org/10.1021/acs.jpcc.2c06706> (2023).
355. Noji, M. et al. Breakdown of supersaturation barrier links protein folding to amyloid formation. *Commun. Biol.* **4**, 120 (2021).
356. Hardy, J. Expression of normal sequence pathogenic proteins for neurodegenerative disease contributes to disease risk: 'permissive templating' as a general mechanism underlying neurodegeneration. *Biochem. Soc. Trans.* **33**, 578–581 (2005).
357. Ciryam, P., Tartaglia, G. G., Morimoto, R. I., Dobson, C. M. & Vendruscolo, M. Widespread aggregation and neurodegenerative diseases are associated with supersaturated proteins. *Cell Rep.* **5**, 781–790 (2013).
358. Ciryam, P. et al. A transcriptional signature of Alzheimer's disease is associated with a metastable subproteome at risk for aggregation. *Proc. Natl Acad. Sci. USA* **113**, 4753–4758 (2016).
359. Freer, R. et al. Supersaturated proteins are enriched at synapses and underlie cell and tissue vulnerability in Alzheimer's disease. *Heliyon* **5**, e02589 (2019).
360. Ciryam, P. et al. Spinal motor neuron protein supersaturation patterns are associated with inclusion body formation in ALS. *Proc. Natl Acad. Sci. USA* **114**, E3935–E3943 (2017).

361. Walther, D. M. et al. Widespread proteome remodeling and aggregation in aging *C. elegans*. *Cell* **161**, 919–932 (2015).
362. Kundra, R., Ciryam, P., Morimoto, R. I., Dobson, C. M. & Vendruscolo, M. Protein homeostasis of a metastable subproteome associated with Alzheimer's disease. *Proc. Natl Acad. Sci. USA* **114**, E5703–E5711 (2017).
363. Huiting, W. et al. Targeting DNA topoisomerases or checkpoint kinases results in an overload of chaperone systems, triggering aggregation of a metastable subproteome. *eLife* **11**, e70726 (2022).
364. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
365. Akdel, M. et al. A structural biology community assessment of AlphaFold2 applications. *Nat. Struct. Mol. Biol.* **29**, 1056–1067 (2022).
366. Ma, P., Li, D.-W. & Brüschweiler, R. Predicting protein flexibility with AlphaFold. *Proteins Struct. Funct. Bioinforma.* **91**, 847–855 (2023).
367. Guo, H.-B. et al. AlphaFold2 models indicate that protein sequence determines both structure and dynamics. *Sci. Rep.* **12**, 10696 (2022).
368. Terwilliger, T. C. et al. Improved AlphaFold modeling with implicit experimental information. *Nat. Methods* **19**, 1376–1382 (2022).
369. Baek, M. et al. Accurate prediction of protein structures and interactions using a three-track neural network. *Science* **373**, 871–876 (2021).
370. Mirdita, M. et al. ColabFold: making protein folding accessible to all. *Nat. Methods* **19**, 679–682 (2022).
371. Wang, W., Peng, Z. & Yang, J. Single-sequence protein structure prediction using supervised transformer protein language models. *Nat. Comput. Sci.* **2**, 804–814 (2022).
372. Richard, E. et al. Protein complex prediction with AlphaFold-Multimer. Preprint at *bioRxiv* <https://doi.org/10.1101/2021.10.04.463034> (2022).
373. Bryant, P., Pozzati, G. & Elofsson, A. Improved prediction of protein–protein interactions using AlphaFold2. *Nat. Commun.* **13**, 1265 (2022).
374. Bryant, P. et al. Predicting the structure of large protein complexes using AlphaFold and Monte Carlo tree search. *Nat. Commun.* **13**, 6028 (2022).
375. Hekkelman, M. L., de Vries, I., Joosten, R. P. & Perrakis, A. AlphaFill: enriching AlphaFold models with ligands and cofactors. *Nat. Methods* **20**, 205–213 (2023).
376. Minkyung, B., Ryan, M., Ivan, A., David, B. & Frank, D. Accurate prediction of nucleic acid and protein–nucleic acid complexes using RoseTTAFoldNA. Preprint at *bioRxiv* <https://doi.org/10.1101/2022.09.09.507333> (2022).
377. Madani, A. et al. Large language models generate functional protein sequences across diverse families. *Nat. Biotechnol.* <https://doi.org/10.1038/s41587-022-01618-2> (2023).
378. Pinheiro, F., Santos, J. & Ventura, S. AlphaFold and the amyloid landscape. *J. Mol. Biol.* **433**, 167059 (2021).
379. Wojciechowski, J. W. & Kotulska, M. PATH—prediction of amyloidogenicity by threading and machine learning. *Sci. Rep.* **10**, 7721 (2020).
380. Navarro, S. & Ventura, S. Computational methods to predict protein aggregation. *Curr. Opin. Struct. Biol.* **73**, 102343 (2022).
381. Meng, F. & Chung, H. S. Kinetics of amyloid β from deep learning. *Nat. Comput. Sci.* **1**, 20–21 (2021).
382. Huda, A. et al. A machine learning model for identifying patients at risk for wild-type transthyretin amyloid cardiomyopathy. *Nat. Commun.* **12**, 2725 (2021).
383. Kim, N. H. et al. PET-validated EEG-machine learning algorithm predicts brain amyloid pathology in pre-dementia Alzheimer's disease. *Sci. Rep.* **13**, 10299 (2023).
384. Tang, Z. et al. Interpretable classification of Alzheimer's disease pathologies with a convolutional neural network pipeline. *Nat. Commun.* **10**, 2173 (2019).
385. Zozulia, O., Dolan, M. A. & Korendovych, I. V. Catalytic peptide assemblies. *Chem. Soc. Rev.* **47**, 3621–3639 (2018).
386. Li, J. & Zhang, F. Amyloids as building blocks for macroscopic functional materials: designs, applications and challenges. *Int. J. Mol. Sci.* <https://doi.org/10.3390/ijms221910698> (2021).
387. Li, J. et al. Microbially synthesized polymeric amyloid fiber promotes β -nanocrystal formation and displays gigapascal tensile strength. *ACS Nano* **15**, 11843–11853 (2021).
388. Sackewitz, M. et al. A folded and functional protein domain in an amyloid-like fibril. *Protein Sci.* **17**, 1044–1054 (2008).
389. Pilkington, S. M., Roberts, S. J., Meade, S. J. & Gerrard, J. A. Amyloid fibrils as a nanoscaffold for enzyme immobilization. *Biotechnol. Prog.* **26**, 93–100 (2010).
390. Goswami, S., Reja, A., Pal, S., Singh, A. & Das, D. Nonequilibrium amyloid polymers exploit dynamic covalent linkage to temporally control charge-selective catalysis. *J. Am. Chem. Soc.* **144**, 19248–19252 (2022).
391. Zozulia, O., Marshall, L. R., Kim, I., Kohn, E. M. & Korendovych, I. V. Self-assembling catalytic peptide nanomaterials capable of highly efficient peroxidase activity. *Chemistry* **27**, 5388–5392 (2021).
392. Zozulia, O. & Korendovych, I. V. Semi-rationally designed short peptides self-assemble and bind hemin to promote cyclopropanation. *Angew. Chem. Int. Ed. Engl.* **59**, 8108–8112 (2020).
393. Sarkhel, B., Chatterjee, A. & Das, D. Covalent catalysis by cross β amyloid nanotubes. *J. Am. Chem. Soc.* **142**, 4098–4103 (2020).
394. Heerde, T., Bansal, A., Schmidt, M. & Fandrich, M. Cryo-EM structure of a catalytic amyloid fibril. *Sci. Rep.* **13**, 4070 (2023).
395. Huyst, A. M. R. et al. Impact of heat and enzymatic treatment on ovalbumin amyloid-like fibril formation and enzyme-induced gelation. *Food Hydrocoll.* **131**, 107784 (2022).
396. Monge-Morera, M. et al. Processing induced changes in food proteins: amyloid formation during boiling of hen egg white. *Biomacromolecules* **21**, 2218–2228 (2020).
397. Jansens, K. J. A. et al. Rational design of amyloid-like fibrillary structures for tailoring food protein techno-functionality and their potential health implications. *Compr. Rev. Food Sci. F.* **18**, 84–105 (2019).
398. Luyckx, T. et al. Bioavailability and health impact of ingested amyloid-like protein fibrils and their link with inflammatory status: a need for more research? *Mol. Nutr. Food Res.* **66**, e2101032 (2022).
399. Ganesan, A. et al. Selectivity of aggregation-determining interactions. *J. Mol. Biol.* **427**, 236–247 (2015).
400. Wu, G. et al. Investigating the mechanism of action of aggregation-inducing antimicrobial Pept-ins. *Cell Chem. Biol.* **28**, 524–536.e4 (2021).
401. Bednarska, N. G. et al. Protein aggregation as an antibiotic design strategy. *Mol. Microbiol.* <https://doi.org/10.1111/mmi.13269> (2015).
402. Siemons, M. et al. Synthetic Pept-ins as a generic amyloid-like aggregation-based platform for in vivo PET imaging of intracellular targets. *Bioconjug. Chem.* **32**, 2052–2064 (2021).

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All authors researched data for the article, made a substantial contribution to discussion of content, and wrote and reviewed/edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

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